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Systems and methods to optimize warehouse operations

Abstract

A system configured to: receive a request to generate a dashboard visualization associated with one or more edge devices, the request comprising: a categorization of one or more of the edge devices; and at least one operational descriptor; and in response to the request: obtain, based on the categorization of the one or more edge devices and the operational descriptor, aggregated data associated with the one or more edge devices; determine a status associated with the one or more edge devices for the aggregated data; provide the dashboard visualization to an electronic interface of a computing device, the dashboard visualization comprising the status associated with the one or more edge devices, determine a list of prioritized actions for the one or more edge devices based on the status; and optimize a status of the one or more edge devices based on the determined list of prioritized actions.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority under 35 U.S.C. § 119 from U.S. Provisional Patent Application No. 63/366,220, filed on Jun. 10, 2022, the contents of which are incorporated by reference in their entirety.

TECHNICAL FIELD

(1) The present disclosure relates generally to methods and systems to optimize operations in a

workplace such as a warehouse or distribution center.

BACKGROUND

(2) Warehouses and distribution centers where employees are often engaged in a multitude of tasks can benefit from receiving real time and historical data from other sources. Further, overall operations can benefit from transmitting real time and historical data to optimize employee operations. Data patterns and trends can be determined from the received data, and the recipient can utilize the data patterns and trends to perform meaningful actions. In practice, employee task optimization is often lacking since a significant amount of optimization benefits have remained unreachable. Therefore, there is a need for a system for collecting and analyzing real-time data from employees, and also for sharing critical data through a streamlined communication network.

(3) The background description provided herein is for the purpose of generally presenting the context of the disclosure. Unless otherwise indicated herein, the materials described in this section are not prior art to the claims in this application and are not admitted to be prior art, or suggestions of the prior art, by inclusion in this section.

SUMMARY

(4) In one embodiment, a system for monitoring performance of a warehouse, includes one or more processors; a memory, comprising one or more programs stored in the memory, the one or more programs comprising instructions configured to: receive a request to generate a dashboard visualization associated with one or more edge devices, the request comprising: a categorization of one or more of the edge devices; and at least one operational descriptor; and in response to the request: obtain, based on the categorization of the one or more edge devices and the operational descriptor, aggregated data associated with the one or more edge devices; determine a status associated with the one or more edge devices for the aggregated data; provide the dashboard visualization to an electronic interface of a computing device, the dashboard visualization comprising the status associated with the one or more edge devices, determine a list of prioritized actions for the one or more edge devices based on the status; and optimize a status of the one or more edge devices based on the determined list of prioritized actions.

(5) In another embodiment, a method of monitoring performance of one or more a warehouses includes receiving a request to generate a dashboard visualization associated with one or more edge devices, the request comprising: a categorization of one or more of the edge devices; and at least one operational descriptor; and in response to the request; obtain, based on the categorization of the one or more edge devices and the operational descriptor, aggregated data associated with the one or more edge devices; determine a status associated with the one or more edge devices for the aggregated data; and provide the dashboard visualization to an electronic interface of a computing device, the dashboard visualization comprising the status associated with the one or more edge devices.

(6) In yet another embodiment, a system includes at least one warehouse, each at least one warehouse including a plurality of edge devices configured to access a network; and a computing device including: a display; one or more processors; a memory including one or more programs stored therein, which, when executed by the one or more processors, cause the computing device to: receive, from the plurality of edge devices, aggregated data associated with the plurality of edge devices; determine prioritized actions associated with the plurality of edge devices for the aggregated data; and provide a dashboard visualization to the display, the dashboard visualization comprising the prioritized actions associated with a portfolio of assets, determine a list of prioritized actions for the one or more edge devices based on the status; and optimize a status of the one or more edge devices based on the determined list of prioritized actions, wherein the visualization is based on one or more of: a categorization of one or more of the edge devices; and at least one operational descriptor.

(7) To the accomplishment of the foregoing and related ends, certain illustrative aspects are described herein in connection with the following description and the appended drawings. These

aspects are indicative, however, of but a few of the various ways in which the principles of the claimed subject matter may be employed and the claimed subject matter is intended to include all such aspects and their equivalents. Other advantages and novel features may become apparent from the following detailed description when considered in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments of the disclosure will now be described, by way of example only, with reference to the accompanying drawings in which:
- (2) FIG. 1 illustrates an exemplary networked computing system environment, in accordance with one or more embodiments described herein;
- (3) FIG. 2 illustrates a schematic block diagram of a framework of an IoT platform of the networked computing system, in accordance with one or more embodiments described herein;
- (4) FIG. 3 illustrates a system that provides an exemplary environment, in accordance with one or more embodiments described herein;
- (5) FIG. 4 illustrates an exemplary computing device, in accordance with one or more embodiments described herein;
- (6) FIG. 5 illustrates an exemplary warehouse environment implementing methods and systems of this disclosure;
- (7) FIG. 6 illustrates an exemplary site operations manager who may implement methods and systems of this disclosure;
- (8) FIG. 7 illustrates an exemplary shift supervisor who may implement methods and systems of this disclosure;
- (9) FIG. 8 illustrates an exemplary user interface dashboard in a first mode, in accordance with one or more embodiments described herein;
- (10) FIG. 9 illustrates an exemplary dashboard for displaying information including recommendations to a user, in accordance with one or more embodiments described herein;
- (11) FIG. 10 illustrates another exemplary electronic interface, in accordance with one or more embodiments described herein;
- (12) FIG. 11 illustrates a schematic view of a material handling system including LiDAR based vision system, in accordance with one or more embodiments described herein;
- (13) FIG. 12 illustrates a schematic view of a target area of the material handling system including the LiDAR based vision system, in accordance with one or more embodiments described herein;
- (14) FIG. 13 illustrates an example scenario depicting monitoring of an operation performed by a worker in a material handling environment by using LiDAR based vision system, in accordance with one or more embodiments described herein;
- (15) FIG. 14 illustrates another example scenario depicting another operation performed in a material handling environment that can be monitored by using LiDAR based vision system, in accordance with one or more embodiments described herein;
- (16) FIG. 15 illustrates a functional block diagram of a computer that may be configured to execute techniques described in accordance with one or more embodiments described herein.

DETAILED DESCRIPTION

- (17) The following embodiments describe systems and methods for facilitating a connected warehouse as between employees, managers, and other users as well as inter- and intra-warehouse edge communications systems.
- (18) Current tools for monitoring warehouse operations may require users to extract data and generate insights from separate systems with disconnected interfaces and non-communicating components. Disaggregated, stovepiped systems must be routinely monitored and the data they

create must be analyzed separately by users and then aggregated manually in order to fully optimize warehouse operations. This disjointed system architecture can lead to inefficiencies and overall suboptimal warehouse operations.

(19) A dynamic and decentralized system and methods for implementing a connected warehouse system is provided. An embodiment or implementation described herein as “dynamic” is intended to reflect or indicate that the embodiment(s) is or can be marked by continuous and productive activity or change, though not necessarily constantly changing. The system and corresponding techniques facilitate communications within a warehouse, between users (e.g., worker, teams of workers, manager, etc.), and between warehouses, third parties associated therewith, and data centers. Such communications may be facilitated by edge systems and gateway systems. The edge and gateway systems may be located in warehouses (i.e., on-site) as embedded or fixed systems and/or other user devices such as tablet PCs and mobile phones. Each edge system may be coupled to a warehouse system from which warehouse operations data may be collected, and in communication with other edge systems and gateway systems. Each gateway system may be in communication with warehouse operation systems and edge systems of the warehouse in which the gateway system is resident, and may also be in communication with gateway systems located in other warehouses, all or some of which may provide data to the gateway system. By facilitating communication with gateway systems located in other warehouses, the gateway system may enable exchange of data among edge systems installed in different warehouses. Independent user computing devices, such as tablet PCs and mobile phones, may be directly coupled to and/or in communication with the edge systems and/or gateway systems, to request, filter, view, and/or analyze data.

(20) Hardware for all or some of the edge systems and gateway systems may be installed in warehouses. Therefore, software may be installed on the corresponding warehouse hardware. The software implemented in the edge systems and gateway systems may comprise computer-executable code for performing various data functions, including but not limited to, data request, data query, data retrieval, data transmission, and data analytics. The edge systems and gateway systems each identify source(s) of relevant data, and request that data be provided dynamically (as needed) or statically (all the time) from the identified source(s), such as from other edge systems coupled to warehouse systems in the warehouse or other warehouses, gateway systems in the warehouse or other warehouses, decentralized system(s) such as cloud computing center(s), and centralized system(s) such as dedicated server farms. The decentralized system(s) and centralized system(s) may be owned by the operators of the warehouses, or by a third party such as a government or a commercial entity.

(21) Each edge system in a warehouse may be coupled to a sensor of a corresponding warehouse system in the same warehouse, enabling data captured by the sensor to be provided directly to the edge system. Also, a gateway system in a warehouse may be coupled to one or more sensors of warehouse systems in the same warehouse, enabling data captured by the one or more sensors to be provided directly to the gateway system. In another embodiment, each edge system in a warehouse may be coupled to warehouse system of a corresponding warehouse system in the same warehouse. Also, a gateway system in a warehouse may be coupled to warehouse system machines of warehouse systems in the same warehouse. In some aspects, warehouse system machines may be configured to collect data from the coupled one or more sensors, perform computations and/or analysis of the collected data, store the collected and/or analyzed data in memory, and provide the collected and/or analyzed data to one or more connected edge systems and/or gateway system. In some embodiments, the warehouse system may not be implemented, or may not be coupled to the one or more sensors of the warehouse system. If the warehouse system machine is not implemented or not coupled to the one or more sensors, data captured by the one or more sensors may be provided directly to the one or more connected edge systems and/or gateway system.

(22) Each warehouse system may be in communication with, through an edge system or not, a

gateway system. Edge systems in a warehouse may be in direct communication with one another. For example, any data retained by one edge system may be transmitted directly to another edge system within the same warehouse, without a gateway system acting as an intermediary. In another embodiment, an edge system may send to or receive data from another edge system located in the same warehouse through a gateway system. The communication between the edge systems and the communication between the edge systems and the gateway system may be through a wired or wireless connection.

(23) A gateway system of a warehouse may be in communication with gateway systems of other warehouses. Through this communication path, an edge system or a gateway system of a warehouse may transmit data to and obtain data from edge systems or gateway systems of other warehouses. The communication path between gateway systems of different warehouses may be through satellite communications (e.g., SATCOM), cellular networks, Wi-Fi (e.g., IEEE 802.11 compliant), WiMAX (e.g., AeroMACS), optical fiber, and/or air-to-ground (ATG) network, and/or any other communication links now known or later developed. An edge system in a warehouse may communicate with another edge system in a different warehouse via gateway systems of the respective warehouses. For example, an edge system in a warehouse may transmit data to one or more edge systems in other warehouses via the gateway systems of the respective warehouses communicating over the communication path discussed above.

(24) Each edge system and gateway system may comprise state machines, such as processor(s) coupled to memory. Both the edge systems and the gateway systems may be configured with a common operating system to support portable, system-wide edge software implementations. In other words, each of the edge systems and the gateway systems may be equipped with standard software to facilitate inter-operability among the edge systems and the gateway systems. In the discussion below, such software will be referred to as edge software. The edge software may enable each edge system or gateway system to perform various functions listed below (non-exhaustive) to enable data analysis and data exchange among the various systems illustrated herein (e.g., edge systems, gateway systems, warehouse operations centers, remote systems): Filter and analyze real-time and stored data collected from other edge systems, warehouse systems, gateway systems, and/or operations center(s), and generate events based on the analysis; Identify dynamic (i.e., as needed) and static (i.e., all the time) data transmission targets (e.g., edge systems within the same warehouse, edge systems in other warehouses, operations center(s)); Transmit data over an Internet connection to the operations centers; Transmit data and events to other edge and gateway systems within an aircraft that are connected over wired/wireless networks, or to other edge and gateway systems external to the aircraft that are connected over the Internet; Provide a request/response interface for other edge/gateway systems, warehouse borne computer systems, operations centers, and remote systems connected over wired/wireless networks or Internet to query the stored data and to dynamically select/change data filters; Use request/response interfaces provided by other edge systems, gateway systems, and operations centers connected over wired/wireless networks or Internet to obtain data and to dynamically select/change data filters; Receive events from other edge systems, gateway systems, and operations centers; and Specify and communicate generic purposes (i.e., types of data the edge/gateway system is interested in) to other edge systems, gateway systems, and operations centers.

(25) Each edge system or gateway system may autonomously select and deliver data to one or more transmission targets, which may be other edge systems in the same warehouse, edge systems in other warehouses, gateway system in the same warehouse, gateway systems in other warehouses, or operations center(s). Each of the receiving edge or gateway systems (i.e., transmission targets) may be configured to filter the received data using a pre-defined filter, overriding the autonomous determination made by the edge system transmitting the data. In some embodiment, each receiving edge or gateway system may notify the other systems, in advance of the data transmission, of the types of data and/or analysis the receiving system wants to receive (i.e., generic “purposes”). Also,

each edge or gateway system may maintain a list including static data transmission targets (transmission targets that always need the data) and dynamic data transmission targets (transmission targets that need the data on as-needed basis).

(26) A gateway system of a warehouse may also be in communication with one or more operations centers, which may be located remotely from the warehouse (i.e., off-site). In some embodiments, however, the operations center(s) may be located on-site at the warehouse. Each of the warehouse systems of this disclosure may be implemented in a dedicated location, such as a server system, or may be implemented in a decentralized manner, for example, as part of a cloud system. The communication path between the gateway systems and the operations center(s) may be through satellite communications (e.g., SATCOM), cellular networks, Wi-Fi (e.g., IEEE 802.11 compliant), WiMAX (e.g., AeroMACS), optical fiber, and/or air-to-ground (ATG) network, and/or any other communication links now known or later developed.

(27) Subject matter will now be described more fully hereinafter with reference to the accompanying drawings, which form a part hereof, and which show, by way of illustration, specific exemplary embodiments. An embodiment or implementation described herein as “exemplary” is not to be construed as preferred or advantageous, for example, over other embodiments or implementations; rather, it is intended reflect or indicate that the embodiment(s) is/are “example” embodiment(s). Subject matter be embodied in a variety of different forms and, therefore, covered or claimed subject matter is intended to be construed as not being limited to any exemplary embodiments set forth herein; exemplary embodiments are provided merely to be illustrative. Likewise, a reasonably broad scope for claimed or covered subject matter is intended. Among other things, for example, subject matter may be embodied as methods, devices, components, or systems. Accordingly, embodiments may, for example, take the form of hardware, software, firmware or any combination thereof (other than software per se). Furthermore, the method presented in the drawings and the specification is not to be construed as limiting the order in which the individual steps may be performed. The following detailed description is, therefore, not intended to be taken in a limiting sense.

(28) Throughout the specification and claims, terms may have nuanced meanings suggested or implied in context beyond an explicitly stated meaning. Likewise, the phrase “in one embodiment” as used herein does not necessarily refer to the same embodiment and the phrase “in another embodiment” or “in some embodiments” as used herein does not necessarily refer to a different embodiment. It is intended, for example, that claimed subject matter include combinations of exemplary embodiments in whole or in part.

(29) FIG. 1 illustrates an exemplary networked computing system environment **100**, according to the present disclosure. As shown in FIG. 1, networked computing system environment **100** is organized into a plurality of layers including a cloud **105**, a network **110**, and an edge **115**. As detailed further below, components of the edge **115** are in communication with components of the cloud **105** via network **110**.

(30) In various embodiments, network **110** is any suitable network or combination of networks and supports any appropriate protocol suitable for communication of data to and from components of the cloud **105** and between various other components in the networked computing system environment **100** (e.g., components of the edge **115**). According to various embodiments, network **110** includes a public network (e.g., the Internet), a private network (e.g., a network within an organization), or a combination of public and/or private networks. According to various embodiments, network **110** is configured to provide communication between various components depicted in FIG. 1. According to various embodiments, network **110** comprises one or more networks that connect devices and/or components in the network layout to allow communication between the devices and/or components. For example, in one or more embodiments, the network **110** is implemented as the Internet, a wireless network, a wired network (e.g., Ethernet), a local area network (LAN), a Wide Area Network (WANs), Bluetooth, Near Field Communication (NFC),

or any other type of network that provides communications between one or more components of the network layout. In some embodiments, network **110** is implemented using cellular networks, satellite, licensed radio, or a combination of cellular, satellite, licensed radio, and/or unlicensed radio networks.

(31) Components of the cloud **105** include one or more computer systems **120** that form a so-called “Internet-of-Things” or “IoT” platform **125**. It should be appreciated that “IoT platform” is an optional term describing a platform connecting any type of Internet-connected device, and should not be construed as limiting on the types of computing systems useable within IoT platform **125**. In particular, in various embodiments, computer systems **120** includes any type or quantity of one or more processors and one or more data storage devices comprising memory for storing and executing applications or software modules of networked computing system environment **100**. In one embodiment, the processors and data storage devices are embodied in server-class hardware, such as enterprise-level servers. For example, in an embodiment, the processors and data storage devices comprise any type or combination of application servers, communication servers, web servers, super-computing servers, database servers, file servers, mail servers, proxy servers, and/virtual servers. Further, the one or more processors are configured to access the memory and execute processor-readable instructions, which when executed by the processors configures the processors to perform a plurality of functions of the networked computing system environment **100**.

(32) Computer systems **120** further include one or more software components of the IoT platform **125**. For example, in one or more embodiments, the software components of computer systems **120** include one or more software modules to communicate with user devices and/or other computing devices through network **110**. For example, in one or more embodiments, the software components include one or more modules **141**, models **142**, engines **143**, databases **144**, services **145**, and/or applications **146**, which may be stored in/by the computer systems **120** (e.g., stored on the memory), as detailed with respect to FIG. 2 below. According to various embodiments, the one or more processors are configured to utilize the one or more modules **141**, models **142**, engines **143**, databases **144**, services **145**, and/or applications **146** when performing various methods described in this disclosure.

(33) Accordingly, in one or more embodiments, computer systems **120** execute a cloud computing platform (e.g., IoT platform **125**) with scalable resources for computation and/or data storage, and may run one or more applications on the cloud computing platform to perform various computer-implemented methods described in this disclosure. In some embodiments, some of the modules **141**, models **142**, engines **143**, databases **144**, services **145**, and/or applications **146** are combined to form fewer modules, models, engines, databases, services, and/or applications. In some embodiments, some of the modules **141**, models **142**, engines **143**, databases **144**, services **145**, and/or applications **146** are separated into separate, more numerous modules, models, engines, databases, services, and/or applications. In some embodiments, some of the modules **141**, models **142**, engines **143**, databases **144**, services **145**, and/or applications **146** are removed while others are added.

(34) The computer systems **120** are configured to receive data from other components (e.g., components of the edge **115**) of networked computing system environment **100** via network **110**. Computer systems **120** are further configured to utilize the received data to produce a result. According to various embodiments, information indicating the result is transmitted to users via user computing devices over network **110**. In some embodiments, the computer systems **120** is a server system that provides one or more services including providing the information indicating the received data and/or the result(s) to the users. According to various embodiments, computer systems **120** are part of an entity which include any type of company, organization, or institution that implements one or more IoT services. In some examples, the entity is an IoT platform provider.

(35) Components of the edge **115** include one or more enterprises **160a-160n** each including one or

more edge devices **161a-161n** and one or more edge gateways **162a-162n**. For example, a first enterprise **160a** includes first edge devices **161a** and first edge gateways **162a**, a second enterprise **160b** includes second edge devices **161b** and second edge gateways **162b**, and an nth enterprise **160n** includes nth edge devices **161n** and nth edge gateways **162n**. As used herein, enterprises **160a-160n** represent any type of entity, facility, or vehicle, such as, for example, companies, divisions, buildings, manufacturing plants, warehouses, real estate facilities, laboratories, aircraft, spacecraft, automobiles, ships, boats, military vehicles, oil and gas facilities, or any other type of entity, facility, and/or entity that includes any number of local devices.

(36) According to various embodiments, the edge devices **161a-161n** represent any of a variety of different types or categories of devices that may be found within the enterprises **160a-160n**. Edge devices **161a-161n** are any type of device configured to access network **110**, or be accessed by other devices through network **110**, such as via an edge gateway **162a-162n**. According to various embodiments, edge devices **161a-161n** are “IoT devices” which include any type of network-connected (e.g., Internet-connected) device. For example, in one or more embodiments, the edge devices **161a-161n** include assets, sensors, actuators, processors, computers, valves, pumps, ducts, vehicle components, cameras, displays, doors, windows, security components, boilers, chillers, pumps, HVAC components, factory equipment, and/or any other devices that are connected to the network **110** for collecting, sending, and/or receiving information. Each edge device **161a-161n** includes, or is otherwise in communication with, one or more controllers for selectively controlling a respective edge device **161a-161n** and/or for sending/receiving information between the edge devices **161a-161n** and the cloud **105** via network **110**. A categorization of the edge device **161a-161n** may be used as input to a dashboard visualization to provide information and/or recommendations to a user of the systems described herein.

(37) With reference to FIG. 2, in one or more embodiments, the edge **115** include operational technology (OT) systems **163a-163n** and information technology (IT) applications **164a-164n** of each enterprise **161a-161n**. The OT systems **163a-163n** include hardware and software for detecting and/or causing a change, through the direct monitoring and/or control of industrial equipment (e.g., edge devices **161a-161n**), assets, processes, and/or events. The IT applications **164a-164n** includes network, storage, and computing resources for the generation, management, storage, and delivery of data throughout and between organizations.

(38) The edge gateways **162a-162n** include devices for facilitating communication between the edge devices **161a-161n** and the cloud **105** via network **110**. For example, the edge gateways **162a-162n** include one or more communication interfaces for communicating with the edge devices **161a-161n** and for communicating with the cloud **105** via network **110**. According to various embodiments, the communication interfaces of the edge gateways **162a-162n** include one or more cellular radios, Bluetooth, WiFi, near-field communication radios, Ethernet, or other appropriate communication devices for transmitting and receiving information. According to various embodiments, multiple communication interfaces are included in each gateway **162a-162n** for providing multiple forms of communication between the edge devices **161a-161n**, the gateways **162a-162n**, and the cloud **105** via network **110**. For example, in one or more embodiments, communication are achieved with the edge devices **161a-161n** and/or the network **110** through wireless communication (e.g., WiFi, radio communication, etc.) and/or a wired data connection (e.g., a universal serial bus, an onboard diagnostic system, etc.) or other communication modes, such as a local area network (LAN), wide area network (WAN) such as the Internet, a telecommunications network, a data network, or any other type of network.

(39) According to various embodiments, the edge gateways **162a-162n** also include a processor and memory for storing and executing program instructions to facilitate data processing. For example, in one or more embodiments, the edge gateways **162a-162n** are configured to receive data from the edge devices **161a-161n** and process the data prior to sending the data to the cloud **105**.

Accordingly, in one or more embodiments, the edge gateways **162a-162n** include one or more

software modules or components for providing data processing services and/or other services or methods of the present disclosure. With reference to FIG. 2, each edge gateway **162a-162n** includes edge services **165a-165n** and edge connectors **166a-166n**. According to various embodiments, the edge services **165a-165n** include hardware and software components for processing the data from the edge devices **161a-161n**. According to various embodiments, the edge connectors **166a-166n** include hardware and software components for facilitating communication between the edge gateway **162a-162n** and the cloud **105** via network **110**, as detailed above. In some cases, any of edge devices **161a-n**, edge connectors **166a-n**, and edge gateways **162a-n** have their functionality combined, omitted, or separated into any combination of devices. In other words, an edge device and its connector and gateway need not necessarily be discrete devices.

(40) FIG. 2 illustrates a schematic block diagram of framework **200** of the IoT platform **125**, according to the present disclosure. The IoT platform **125** of the present disclosure is a platform for enterprise performance management that uses real-time accurate models and visual analytics to deliver intelligent actionable recommendations and/or analytics for sustained peak performance of the enterprise **160a-160n**, for example, to a regional manager of the enterprise. The IoT platform **125** is an extensible platform that is portable for deployment in any cloud or data center environment for providing an enterprise-wide, top to bottom view, displaying the status of processes, assets, people, and safety. Further, the IoT platform **125** supports end-to-end capability to execute digital twins against process data and to translate the output into actionable insights, using the framework **200**, detailed further below.

(41) As shown in FIG. 2, the framework **200** of the IoT platform **125** comprises a number of layers including, for example, an IoT layer **205**, an enterprise integration layer **210**, a data pipeline layer **215**, a data insight layer **220**, an application services layer **225**, and an applications layer **230**. The IoT platform **125** also includes a core services layer **235** and an extensible object model (EOM) **250** comprising one or more knowledge graphs **251**. The layers **205-235** further include various software components that together form each layer **205-235**. For example, in one or more embodiments, each layer **205-235** includes one or more of the modules **141**, models **142**, engines **143**, databases **144**, services **145**, applications **146**, or combinations thereof. In some embodiments, the layers **205-235** are combined to form fewer layers. In some embodiments, some of the layers **205-235** are separated into separate, more numerous layers. In some embodiments, some of the layers **205-235** are removed while others may be added.

(42) The IoT platform **125** is a model-driven architecture. Thus, the extensible object model **250** communicates with each layer **205-230** to contextualize site data of the enterprise **160a-160n** using an extensible graph based object model (or “asset model”). In one or more embodiments, the extensible object model **250** is associated with knowledge graphs **251** where the equipment (e.g., edge devices **161a-161n**) and processes of the enterprise **160a-160n** are modeled. The knowledge graphs **251** of EOM **250** are configured to store the models in a central location. The knowledge graphs **251** define a collection of nodes and links that describe real-world connections that enable smart systems. As used herein, a knowledge graph **251**: (i) describes real-world entities (e.g., edge devices **161a-161n**) and their interrelations organized in a graphical interface; (ii) defines possible classes and relations of entities in a schema; (iii) enables interrelating arbitrary entities with each other; and (iv) covers various topical domains. In other words, the knowledge graphs **251** define large networks of entities (e.g., edge devices **161a-161n**), semantic types of the entities, properties of the entities, and relationships between the entities. Thus, the knowledge graphs **251** describe a network of “things” that are relevant to a specific domain or to an enterprise or organization. Knowledge graphs **251** are not limited to abstract concepts and relations, but can also contain instances of objects, such as, for example, documents and datasets. In some embodiments, the knowledge graphs **251** include resource description framework (RDF) graphs. As used herein, a “RDF graph” is a graph data model that formally describes the semantics, or meaning, of information. The RDF graph also represents metadata (e.g., data that describes data). According to

various embodiments, knowledge graphs **251** also include a semantic object model. The semantic object model is a subset of a knowledge graph **251** that defines semantics for the knowledge graph **251**. For example, the semantic object model defines the schema for the knowledge graph **251**.

(43) As used herein, EOM **250** includes a collection of application programming interfaces (APIs) that enables seeded semantic object models to be extended. For example, the EOM **250** of the present disclosure enables a customer's knowledge graph **251** to be built subject to constraints expressed in the customer's semantic object model. Thus, the knowledge graphs **251** are generated by customers (e.g., enterprises or organizations) to create models of the edge devices **161a-161n** of an enterprise **160a-160n**, and the knowledge graphs **251** are input into the EOM **250** for visualizing the models (e.g., the nodes and links).

(44) The models describe the assets (e.g., the nodes) of an enterprise (e.g., the edge devices **161a-161n**) and describe the relationship of the assets with other components (e.g., the links). The models also describe the schema (e.g., describe what the data is), and therefore the models are self-validating. For example, in one or more embodiments, the model describes the type of sensors mounted on any given asset (e.g., edge device **161a-161n**) and the type of data that is being sensed by each sensor. According to various embodiments, a KPI framework is used to bind properties of the assets in the extensible object model **250** to inputs of the KPI framework. Accordingly, the IoT platform **125** is an extensible, model-driven end-to-end stack including: two-way model sync and secure data exchange between the edge **115** and the cloud **105**, metadata driven data processing (e.g., rules, calculations, and aggregations), and model driven visualizations and applications. As used herein, "extensible" refers to the ability to extend a data model to include new properties/columns/fields, new classes/tables, and new relations. Thus, the IoT platform **125** is extensible with regards to edge devices **161a-161n** and the applications **146** that handle those devices **161a-161n**. For example, when new edge devices **161a-161n** are added to an enterprise **160a-160n** system, the new devices **161a-161n** will automatically appear in the IoT platform **125** so that the corresponding applications **146** understand and use the data from the new devices **161a-161n**.

(45) In some cases, asset templates are used to facilitate configuration of instances of edge devices **161a-161n** in the model using common structures. An asset template defines the typical properties for the edge devices **161a-161n** of a given enterprise **160a-160n** for a certain type of device. For example, an asset template of a pump includes modeling the pump having inlet and outlet pressures, speed, flow, etc. The templates may also include hierarchical or derived types of edge devices **161a-161n** to accommodate variations of a base type of device **161a-161n**. For example, a reciprocating pump is a specialization of a base pump type and would include additional properties in the template. Instances of the edge device **161a-161n** in the model are configured to match the actual, physical devices of the enterprise **160a-160n** using the templates to define expected attributes of the device **161a-161n**. Each attribute is configured either as a static value (e.g., capacity is 1000 BPH) or with a reference to a time series tag that provides the value. The knowledge graph **251** can automatically map the tag to the attribute based on naming conventions, parsing, and matching the tag and attribute descriptions and/or by comparing the behavior of the time series data with expected behavior. In one or more embodiments, each of the key attributes contributing to one or more metrics to drive a dashboard is marked with one or more metric tags such that a dashboard visualization is generated.

(46) The modeling phase includes an onboarding process for syncing the models between the edge **115** and the cloud **105**. For example, in one or more embodiments, the onboarding process includes a simple onboarding process, a complex onboarding process, and/or a standardized rollout process. The simple onboarding process includes the knowledge graph **251** receiving raw model data from the edge **115** and running context discovery algorithms to generate the model. The context discovery algorithms read the context of the edge naming conventions of the edge devices **161a-161n** and determine what the naming conventions refer to. For example, in one or more

embodiments, the knowledge graph **251** receives “TMP” during the modeling phase and determine that “TMP” relates to “temperature.” The generated models are then published. The complex onboarding process includes the knowledge graph **251** receiving the raw model data, receiving point history data, and receiving site survey data. According to various embodiments, the knowledge graph **251** then uses these inputs to run the context discovery algorithms. According to various embodiments, the generated models are edited and then the models are published. The standardized rollout process includes manually defining standard models in the cloud **105** and pushing the models to the edge **115**.

(47) The IoT layer **205** includes one or more components for device management, data ingest, and/or command/control of the edge devices **161a-161n**. The components of the IoT layer **205** enable data to be ingested into, or otherwise received at, the IoT platform **125** from a variety of sources. For example, in one or more embodiments, data is ingested from the edge devices **161a-161n** through process historians or laboratory information management systems. The IoT layer **205** is in communication with the edge connectors **165a-165n** installed on the edge gateways **162a-162n** through network **110**, and the edge connectors **165a-165n** send the data securely to the IoT layer **205**. In some embodiments, only authorized data is sent to the IoT platform **125**, and the IoT platform **125** only accepts data from authorized edge gateways **162a-162n** and/or edge devices **161a-161n**. According to various embodiments, data is sent from the edge gateways **162a-162n** to the IoT platform **125** via direct streaming and/or via batch delivery. Further, after any network or system outage, data transfer will resume once communication is re-established and any data missed during the outage will be backfilled from the source system or from a cache of the IoT platform **125**. According to various embodiments, the IoT layer **205** also includes components for accessing time series, alarms and events, and transactional data via a variety of protocols.

(48) The enterprise integration layer **210** includes one or more components for events/messaging, file upload, and/or REST/OData. The components of the enterprise integration layer **210** enable the IoT platform **125** to communicate with third party cloud applications **211**, such as any application(s) operated by an enterprise in relation to its edge devices. For example, the enterprise integration layer **210** connects with enterprise databases, such as guest databases, customer databases, financial databases, patient databases, etc. The enterprise integration layer **210** provides a standard application programming interface (API) to third parties for accessing the IoT platform **125**. The enterprise integration layer **210** also enables the IoT platform **125** to communicate with the OT systems **163a-163n** and IT applications **164a-164n** of the enterprise **160a-160n**. Thus, the enterprise integration layer **210** enables the IoT platform **125** to receive data from the third party cloud applications **211** rather than, or in combination with, receiving the data from the edge devices **161a-161n** directly.

(49) The data pipeline layer **215** includes one or more components for data cleansing/enriching, data transformation, data calculations/aggregations, and/or API for data streams. Accordingly, in one or more embodiments, the data pipeline layer **215** pre-processes and/or performs initial analytics on the received data. The data pipeline layer **215** executes advanced data cleansing routines including, for example, data correction, mass balance reconciliation, data conditioning, component balancing and simulation to ensure the desired information is used as a basis for further processing. The data pipeline layer **215** also provides advanced and fast computation. For example, cleansed data is run through enterprise-specific digital twins. According to various embodiments, the enterprise-specific digital twins include a reliability advisor containing process models to determine the current operation and the fault models to trigger any early detection and determine an appropriate resolution. According to various embodiments, the digital twins also include an optimization advisor that integrates real-time economic data with real-time process data, selects the right feed for a process, and determines optimal process conditions and product yields.

(50) According to various embodiments, the data pipeline layer **215** employs models and templates to define calculations and analytics. Additionally or alternatively, according to various

embodiments, the data pipeline layer **215** employs models and templates to define how the calculations and analytics relate to the assets (e.g., the edge devices **161a-161n**). For example, in an embodiment, a pump template defines pump efficiency calculations such that every time a pump is configured, the standard efficiency calculation is automatically executed for the pump. The calculation model defines the various types of calculations, the type of engine that should run the calculations, the input and output parameters, the preprocessing requirement and prerequisites, the schedule, etc. According to various embodiments, the actual calculation or analytic logic is defined in the template or it may be referenced. Thus, according to various embodiments, the calculation model is employed to describe and control the execution of a variety of different process models. According to various embodiments, calculation templates are linked with the asset templates such that when an asset (e.g., edge device **161a-161n**) instance is created, any associated calculation instances are also created with their input and output parameters linked to the appropriate attributes of the asset (e.g., edge device **161a-161n**).

(51) According to various embodiments, the IoT platform **125** supports a variety of different analytics models including, for example, first principles models, empirical models, engineered models, user-defined models, machine learning models, built-in functions, and/or any other types of analytics models. Fault models and predictive maintenance models will now be described by way of example, but any type of models may be applicable.

(52) Fault models are used to compare current and predicted enterprise **160a-160n** performance to identify issues or opportunities, and the potential causes or drivers of the issues or opportunities. The IoT platform **125** includes rich hierarchical symptom-fault models to identify abnormal conditions and their potential consequences. For example, in one or more embodiments, the IoT platform **125** drill downs from a high-level condition to understand the contributing factors, as well as determining the potential impact a lower level condition may have. There may be multiple fault models for a given enterprise **160a-160n** looking at different aspects such as process, equipment, control, and/or operations. According to various embodiments, each fault model identifies issues and opportunities in their domain, and can also look at the same core problem from a different perspective. According to various embodiments, an overall fault model is layered on top to synthesize the different perspectives from each fault model into an overall assessment of the situation and point to the true root cause.

(53) According to various embodiments, when a fault or opportunity is identified, the IoT platform **125** provides recommendations about an optimal corrective action to take. Initially, the recommendations are based on expert knowledge that has been pre-programmed into the system by process and equipment experts. A recommendation services module presents this information in a consistent way regardless of source, and supports workflows to track, close out, and document the recommendation follow-up. According to various embodiments, the recommendation follow-up is employed to improve the overall knowledge of the system over time as existing recommendations are validated (or not) or new cause and effect relationships are learned by users and/or analytics.

(54) According to various embodiments, the models are used to accurately predict what will occur before it occurs and interpret the status of the installed base. Thus, the IoT platform **125** enables operators to quickly initiate maintenance measures when irregularities occur. According to various embodiments, the digital twin architecture of the IoT platform **125** employs a variety of modeling techniques. According to various embodiments, the modeling techniques include, for example, rigorous models, fault detection and diagnostics (FDD), descriptive models, predictive maintenance, prescriptive maintenance, process optimization, and/or any other modeling technique.

(55) According to various embodiments, the rigorous models are converted from process design simulation. In this manner, process design is integrated with feed conditions and production requirement. Process changes and technology improvement provide business opportunities that enable more effective maintenance schedule and deployment of resources in the context of production needs. The fault detection and diagnostics include generalized rule sets that are

specified based on industry experience and domain knowledge and can be easily incorporated and used working together with equipment models. According to various embodiments, the descriptive models identifies a problem and the predictive models determines possible damage levels and maintenance options. According to various embodiments, the descriptive models include models for defining the operating windows for the edge devices **161a-161n**.

(56) Predictive maintenance includes predictive analytics models developed based on rigorous models and statistic models, such as, for example, principal component analysis (PCA) and partial least square (PLS). According to various embodiments, machine learning methods are applied to train models for fault prediction. According to various embodiments, predictive maintenance leverages FDD-based algorithms to continuously monitor individual control and equipment performance. Predictive modeling is then applied to a selected condition indicator that deteriorates in time. Prescriptive maintenance includes determining an optimal maintenance option and when it should be performed based on actual conditions rather than time-based maintenance schedule. According to various embodiments, prescriptive analysis selects the right solution based on the company's capital, operational, and/or other requirements. Process optimization is determining optimal conditions via adjusting set-points and schedules. The optimized set-points and schedules can be communicated directly to the underlying controllers, which enables automated closing of the loop from analytics to control.

(57) The data insight layer **220** includes one or more components for time series databases (TDSB), relational/document databases, data lakes, blob, files, images, and videos, and/or an API for data query. According to various embodiments, when raw data is received at the IoT platform **125**, the raw data is stored as time series tags or events in warm storage (e.g., in a TSDB) to support interactive queries and to cold storage for archive purposes. According to various embodiments, data is sent to the data lakes for offline analytics development. According to various embodiments, the data pipeline layer **215** accesses the data stored in the databases of the data insight layer **220** to perform analytics, as detailed above.

(58) The application services layer **225** includes one or more components for rules engines, workflow/notifications, KPI framework, insights (e.g., actionable insights), decisions, recommendations, machine learning, and/or an API for application services. The application services layer **225** enables building of applications **146a-d**. The applications layer **230** includes one or more applications **146a-d** of the IoT platform **125**. For example, according to various embodiments, the applications **146a-d** includes a buildings application **146a**, a plants application **146b**, an aero application **146c**, and other enterprise applications **146d**. According to various embodiments, the applications **146** includes general applications **146** for portfolio management, asset management, autonomous control, and/or any other custom applications. According to various embodiments, portfolio management includes the KPI framework and a flexible user interface (UI) builder. According to various embodiments, asset management includes asset performance and asset health. According to various embodiments, autonomous control includes energy optimization and/or predictive maintenance. As detailed above, according to various embodiments, the general applications **146** is extensible such that each application **146** is configurable for the different types of enterprises **160a-160n** (e.g., buildings application **146a**, plants application **146b**, aero application **146c**, and other enterprise applications **146d**).

(59) The applications layer **230** also enables visualization of performance of the enterprise **160a-160n**. For example, dashboards provide a high-level overview with drill downs to support deeper investigations. Recommendation summaries give users prioritized actions to address current or potential issues and opportunities. Data analysis tools support ad hoc data exploration to assist in troubleshooting and process improvement.

(60) The core services layer **235** includes one or more services of the IoT platform **125**. According to various embodiments, the core services layer **235** include data visualization, data analytics tools, security, scaling, and monitoring. According to various embodiments, the core services layer **235**

also include services for tenant provisioning, single login/common portal, self-service admin, UI library/UI tiles, identity/access/entitlements, logging/monitoring, usage metering, API gateway/dev portal, and the IoT platform **125** streams.

(61) FIG. 3 illustrates a system **300** that provides an exemplary environment according to one or more described features of one or more embodiments of the disclosure. According to an embodiment, the system **300** includes an asset performance management computer system **302** to facilitate a practical application of data analytics technology and/or digital transformation technology to provide optimization related to enterprise performance management. In one or more embodiments, the asset performance management computer system **302** facilitates a practical application of metrics modeling and/or dynamic cache storage related to dashboard technology to provide optimization related to enterprise performance management. In one or more embodiments, the asset performance management computer system **302** stores and/or analyzes data that is aggregated from one or more assets and/or one or more data sources associated with an enterprise system (e.g., a building system, an industrial system or another type of enterprise system). In one or more embodiments, the asset performance management computer system **302** facilitates a practical application of a virtual assistant related to dashboard technology to provide optimization related to enterprise performance management. In one or more embodiments, the asset performance management computer system **302** employs artificial intelligence to provide the practical application of a virtual assistant related to dashboard technology to provide optimization related to enterprise performance management.

(62) In an embodiment, the asset performance management computer system **302** is a server system (e.g., a server device) that facilitates a data analytics platform between one or more computing devices, one or more data sources, and/or one or more assets. In one or more embodiments, the asset performance management computer system **302** is a device with one or more processors and a memory. In one or more embodiments, the asset performance management computer system **302** is a computer system from the computer systems **120**. For example, in one or more embodiments, the asset performance management computer system **302** is implemented via the cloud **105**. The asset performance management computer system **302** is also related to one or more technologies, such as, for example, enterprise technologies, connected building technologies, industrial technologies, Internet of Things (IoT) technologies, data analytics technologies, digital transformation technologies, cloud computing technologies, cloud database technologies, server technologies, network technologies, private enterprise network technologies, wireless communication technologies, machine learning technologies, artificial intelligence technologies, digital processing technologies, electronic device technologies, computer technologies, supply chain analytics technologies, aircraft technologies, industrial technologies, cybersecurity technologies, navigation technologies, asset visualization technologies, oil and gas technologies, petrochemical technologies, refinery technologies, process plant technologies, procurement technologies, and/or one or more other technologies.

(63) Moreover, the asset performance management computer system **302** provides an improvement to one or more technologies such as enterprise technologies, connected building technologies, industrial technologies, IoT technologies, data analytics technologies, digital transformation technologies, cloud computing technologies, cloud database technologies, server technologies, network technologies, private enterprise network technologies, wireless communication technologies, machine learning technologies, artificial intelligence technologies, digital processing technologies, electronic device technologies, computer technologies, supply chain analytics technologies, aircraft technologies, industrial technologies, cybersecurity technologies, navigation technologies, asset visualization technologies, oil and gas technologies, petrochemical technologies, refinery technologies, process plant technologies, procurement technologies, and/or one or more other technologies. In an implementation, the asset performance management computer system **302** improves performance of a computing device. For example, in one or more

embodiments, the asset performance management computer system **302** improves processing efficiency of a computing device (e.g., a server), reduces power consumption of a computing device (e.g., a server), improves quality of data provided by a computing device (e.g., a server), etc. (64) The asset performance management computer system **302** includes a data aggregation component **304**, a metrics engine component **306**, a prioritized actions component **326**, a virtual assistant component **336**, and/or a dashboard visualization component **308**. Additionally, in one or more embodiments, the asset performance management computer system **302** includes a processor **310** and/or a memory **312**. In certain embodiments, one or more aspects of the asset performance management computer system **302** (and/or other systems, apparatuses and/or processes disclosed herein) constitute executable instructions embodied within a computer-readable storage medium (e.g., the memory **312**). For instance, in an embodiment, the memory **312** stores computer executable component and/or executable instructions (e.g., program instructions). Furthermore, the processor **310** facilitates execution of the computer executable components and/or the executable instructions (e.g., the program instructions). In an example embodiment, the processor **310** is configured to execute instructions stored in the memory **312** or otherwise accessible to the processor **310**.

(65) The processor **310** is a hardware entity (e.g., physically embodied in circuitry) capable of performing operations according to one or more embodiments of the disclosure. Alternatively, in an embodiment where the processor **310** is embodied as an executor of software instructions, the software instructions configure the processor **310** to perform one or more algorithms and/or operations described herein in response to the software instructions being executed. In an embodiment, the processor **310** is a single core processor, a multi-core processor, multiple processors internal to the asset performance management computer system **302**, a remote processor (e.g., a processor implemented on a server), and/or a virtual machine. In certain embodiments, the processor **310** is in communication with the memory **312**, the data aggregation component **304**, the metrics engine component **306**, the prioritized actions component **326**, the virtual assistant component **336** and/or the dashboard visualization component **308** via a bus to, for example, facilitate transmission of data among the processor **310**, the memory **312**, the data aggregation component **304**, the metrics engine component **306**, the prioritized actions component **326**, the virtual assistant component **336** and/or the dashboard visualization component **308**. The processor **310** may be embodied in a number of different ways and, in certain embodiments, includes one or more processing devices configured to perform independently. Additionally or alternatively, in one or more embodiments, the processor **310** includes one or more processors configured in tandem via a bus to enable independent execution of instructions, pipelining of data, and/or multi-thread execution of instructions.

(66) The memory **312** is non-transitory and includes, for example, one or more volatile memories and/or one or more non-volatile memories. In other words, in one or more embodiments, the memory **312** is an electronic storage device (e.g., a computer-readable storage medium). The memory **312** is configured to store information, data, content, one or more applications, one or more instructions, or the like, to enable the asset performance management computer system **302** to carry out various functions in accordance with one or more embodiments disclosed herein. As used herein in this disclosure, the term “component,” “system,” and the like, is a computer-related entity. For instance, “a component,” “a system,” and the like disclosed herein is either hardware, software, or a combination of hardware and software. As an example, a component is, but is not limited to, a process executed on a processor, a processor, circuitry, an executable component, a thread of instructions, a program, and/or a computer entity.

(67) In an embodiment, the asset performance management computer system **302** (e.g., the data aggregation component **304** of the asset performance management computer system **302**) receives asset data **314** from the edge devices **161a-161n**. In one or more embodiments, the edge devices **161a-161n** are associated with a portfolio of assets. For instance, in one or more embodiments, the

edge devices **161a-161n** include one or more assets in a portfolio of assets. The edge devices **161a-161n** include, in one or more embodiments, one or more databases, one or more assets (e.g., one or more building assets, one or more industrial assets, etc.), one or more IoT devices (e.g., one or more industrial IoT devices), one or more connected building assets, one or more sensors, one or more actuators, one or more processors, one or more computers, one or more valves, one or more pumps (e.g., one or more centrifugal pumps, etc.), one or more motors, one or more compressors, one or more turbines, one or more ducts, one or more heaters, one or more chillers, one or more coolers, one or more boilers, one or more furnaces, one or more heat exchangers, one or more fans, one or more blowers, one or more conveyor belts, one or more vehicle components, one or more cameras, one or more displays, one or more security components, one or more HVAC components, industrial equipment, factory equipment, and/or one or more other devices that are connected to the network **110** for collecting, sending, and/or receiving information. In one or more embodiments, the edge device **161a-161n** include, or is otherwise in communication with, one or more controllers for selectively controlling a respective edge device **161a-161n** and/or for sending/receiving information between the edge devices **161a-161n** and the asset performance management computer system **302** via the network **110**. The asset data **314** includes, for example, industrial data, connected building data, sensor data, real-time data, historical data, event data, process data, location data, and/or other data associated with the edge devices **161a-161n**.

(68) In certain embodiments, at least one edge device from the edge devices **161a-161n** incorporates encryption capabilities to facilitate encryption of one or more portions of the asset data **314**. Additionally, in one or more embodiments, the asset performance management computer system **302** (e.g., the data aggregation component **304** of the asset performance management computer system **302**) receives the asset data **314** via the network **110**. In one or more embodiments, the network **110** is a Wi-Fi network, a Near Field Communications (NFC) network, a Worldwide Interoperability for Microwave Access (WiMAX) network, a personal area network (PAN), a short-range wireless network (e.g., a Bluetooth® network), an infrared wireless (e.g., IrDA) network, an ultra-wideband (UWB) network, an induction wireless transmission network, and/or another type of network. In one or more embodiments, the edge devices **161a-161n** are associated with an industrial environment (e.g., a plant, etc.). Additionally or alternatively, in one or more embodiments, the edge devices **161a-161n** are associated with components of the edge **115** such as, for example, one or more enterprises **160a-160n**.

(69) In one or more embodiments, the data aggregation component **304** aggregates the asset data **314** from the edge devices **161a-161n**. For instance, in one or more embodiments, the data aggregation component **304** aggregates the asset data **314** into a centralized control database **318** configured as a database structure. The centralized control database **318** is a cache memory (e.g., a dynamic cache) that dynamically stores the asset data **314** based on interval of time and/or asset hierarchy level. For instance, in one or more embodiments, the centralized control database **318** stores the asset data **314** for one or more intervals of time (e.g., 1 minute to 12 minutes, 1 hour to 24 hours, 1 day to 31 days, 1 month to 12 months, etc.) and/or for one or more asset hierarchy levels (e.g., asset level, asset zone, building level, building zone, plant level, plant zone, industrial site level, etc.). In a non-limiting embodiment, the centralized control database **318** stores the asset data **314** for a first interval of time (e.g., 1 hour to 24 hours minutes) for a first asset (e.g., a first asset hierarchy level), for a second interval of time (e.g., 1 day to 31 days) for the first asset, and for a third interval of time (e.g., 1 month to 12 months) for the first asset.

(70) In an example embodiment, the centralized control database **318** stores the asset data **314** for the first interval of time (e.g., 1 hour to 24 hours minutes) for all assets in a connected building (e.g., a second asset hierarchy level), for the second interval of time (e.g., 1 day to 31 days) for all the assets in the connected building, and for the third interval of time (e.g., 1 month to 12 months) for the all the assets in the connected building. In the example embodiment, the centralized control database **318** also stores the asset data **314** for the first interval of time (e.g., 1 hour to 24 hours

minutes) for all connected buildings within a particular geographic region (e.g., a third asset hierarchy level), for the second interval of time (e.g., 1 day to 31 days) for all connected buildings within the particular geographic region, and for the third interval of time (e.g., 1 month to 12 months) for all connected buildings within the particular geographic region.

(71) In another example embodiment, the centralized control database **318** stores the asset data **314** for the first interval of time (e.g., 1 hour to 24 hours minutes) for all assets in a plant (e.g., a second asset hierarchy level), for the second interval of time (e.g., 1 day to 31 days) for all the assets in the plant, and for the third interval of time (e.g., 1 month to 12 months) for the all the assets in the plant. In the example embodiment, the centralized control database **318** also stores the asset data **314** for the first interval of time (e.g., 1 hour to 24 hours minutes) for all plants at an industrial site (e.g., a third asset hierarchy level), for the second interval of time (e.g., 1 day to 31 days) for all plants at the industrial site, and for the third interval of time (e.g., 1 month to 12 months) for all plants at the industrial site.

(72) In one or more embodiments, the data aggregation component **304** repeatedly updates data of the centralized control database **318** based on the asset data **314** provided by the edge devices **161a-161n** during the one or more intervals of time associated with the centralized control database **318**. For instance, in one or more embodiments, the data aggregation component **304** stores new data and/or modified data associated with the asset data **314**. In one or more embodiments, the data aggregation component **304** repeatedly scans the edge devices **161a-161n** to determine new data for storage in the centralized control database **318**. In one or more embodiments, the data aggregation component **304** formats one or more portions of the asset data **314**. For instance, in one or more embodiments, the data aggregation component **304** provides a formatted version of the asset data **314** to the centralized control database **318**. In an embodiment, the formatted version of the asset data **314** is formatted with one or more defined formats associated with the one or more intervals of time and/or the one or more asset hierarchy levels. A defined format is, for example, a structure for data fields of the centralized control database **318**. In various embodiments, the formatted version of the asset data **314** is stored in the centralized control database **318**.

(73) In one or more embodiments, the data aggregation component **304** identifies and/or groups data types associated with the asset data **314** based on the one or more intervals of time (e.g., one or more reporting intervals of time) and/or the one or more asset hierarchy levels. In one or more embodiments, the data aggregation component **304** employs batching, concatenation of the asset data **314**, identification of data types, merging of the asset data **314**, grouping of the asset data **314**, reading of the asset data **314** and/or writing of the asset data **314** to facilitate storage of the asset data **314** within the centralized control database **318**. In one or more embodiments, the data aggregation component **304** groups data from the asset data **314** based on corresponding features and/or attributes of the data. In one or more embodiments, the data aggregation component **304** groups data from the asset data **314** based on corresponding identifiers (e.g., a matching asset hierarchy level, a matching asset, a matching connected building, etc.) for the asset data **314**. In one or more embodiments, the data aggregation component **304** employs one or more locality-sensitive hashing techniques to group data from the asset data **314** based on similarity scores and/or calculated distances between different data in the asset data **314**.

(74) In one or more embodiments, the data aggregation component **304** organizes the formatted version of the asset data **314** based on a time series mapping of attributes for the asset data **314**. For instance, in one or more embodiments, the data aggregation component **304** employs a hierarchical data format technique to organize the formatted version of the asset data **314** in the centralized control database **318**. In one or more embodiments, the centralized control database **318** dynamically stores data (e.g., one or more portions of the asset data **314**) based on type of data presented via a dashboard visualization. In one or more embodiments, data (e.g., one or more portions of the asset data **314**) aggregated from the edge devices **161a-161n** is converted into one or more metrics (e.g., a KPI metric, a duty KPI, a duty target KPI) prior to being stored in the

centralized control database **318**. In one or more embodiments, a metric (e.g. a KP metrics) consists of aspect data indicative of an aspect employed in a model to map an attribute to the metric (e.g., an operating power asset type attribute is mapped to a duty aspect, etc.), aggregation data indicative of information related to aggregation across time, rollup data indicative of an aggregate metric of an asset across an asset at one level as well as across a hierarchy asset, low limit data indicative of a low-limit constant derived from a digital twin model in real-time, high limit data indicative of a high-limit constant derived from a digital twin model in real-time, target data indicative of a target constant derived from a digital twin model in real-time, custom calculation data indicative of information related to custom calculations using aggregate data across time or asset, and/or other data related to the metric.

(75) In one or more embodiments, the asset performance management computer system **302** (e.g., the prioritized actions component **326** of the asset performance management computer system **302**) receives a request **320**. In an embodiment, the request **320** is a request to generate a dashboard visualization associated with a portfolio of assets. For instance, in one or more embodiments, the request **320** is a request to generate a dashboard visualization associated with the edge devices **161a-161n** (e.g., the edge devices **161a-161n** included in a portfolio of assets).

(76) In one or more embodiments, the request **320** includes one or more asset descriptors that describe one or more assets in the portfolio of assets. For instance, in one or more embodiments, the request **320** includes one or more asset descriptors that describe the edge devices **161a-161n**. An asset descriptor includes, for example, an asset name, an asset identifier, an asset level and/or other information associated with an asset. Additionally or alternatively, in one or more embodiments, the request **320** includes one or more user identifiers describing a user role for a user associated with access of a dashboard visualization. A user identifier includes, for example, an identifier for a user role name (e.g., a manager, an executive, a maintenance engineer, a process engineer, etc.). Additionally or alternatively, in one or more embodiments, the request **320** includes one or more metrics context identifiers describing context for the metrics. A metrics context identifier includes, for example, an identifier for a plant performance metric, an asset performance metric, a goal (e.g., review production related to one or more assets, etc.). Additionally or alternatively, in one or more embodiments, the request **320** includes one or more time interval identifier describing an interval of time for the metrics. A time interval identifier describes, for example, an interval of time for aggregated data such as hourly, daily, monthly, yearly etc. In one or more embodiments, a time interval identifier is a reporting time identifier describing an interval of time for the metrics.

(77) In one or more embodiments, the request **320** is a voice input. In an embodiment, the voice input includes and/or initiates a request to generate a dashboard visualization associated with the portfolio of assets. For instance, in one or more embodiments, the voice input includes and/or initiates a request to generate a dashboard visualization associated with the edge devices **161a-161n** (e.g., the edge devices **161a-161n** included in a portfolio of assets). In one or more embodiments, the voice input comprises voice input data associated with the request to generate the dashboard visualization. For example, in one or more embodiments, the voice input data associated with the voice input comprises one or more asset insight requests associated with the portfolio of assets. In an embodiment, the one or more asset insight requests include a phrase provided via the voice input data. In another embodiment, the one or more asset insight requests include a question provided via the voice input data. For instance, in an embodiment, a user can speak a phrase or a question via a computing device to provide the voice input data associated with the voice input.

(78) In one or more embodiments, the voice input includes one or more attributes (e.g., asset insight attributes, a metrics context identifier, etc.) associated with the one or more asset insight requests. For instance, in one or more embodiments, the voice input includes, for example, an identifier for a plant performance metric, an asset performance metric indicator, a goal indicator, etc. In an example, for a phrase “What was the production and quality of product A?”, the word

“production” can be a first attribute and the word “quality” can be a second attribute. In one or more embodiments, the voice input additionally or alternatively includes one or more asset descriptors that describe one or more assets in the portfolio of assets. For instance, in one or more embodiments, the voice input additionally or alternatively includes one or more asset descriptors that describe the edge devices **161a-161n**. An asset descriptor includes, for example, an asset name, an asset identifier, an asset level and/or other information associated with an asset. Additionally or alternatively, in one or more embodiments, the voice input includes the one or more user identifiers describing a user role for a user associated with access of a dashboard visualization. Additionally or alternatively, in one or more embodiments, the voice input includes time data describing a time and/or an interval of time for the metrics and/or one or more asset insights.

(79) In one or more embodiments, in response to the request **320**, the metrics engine component **306** determines one or more metrics for an asset hierarchy associated with the portfolio of assets. For instance, in one or more embodiments, the metrics engine component **306** determines one or more metrics for an asset hierarchy associated with the edge devices **161a-161n** in response to the request **320**. In one or more embodiments, the metrics engine component **306** converts a portion of the asset data **314** into a metric for the portion of the asset data **314** and stores the metric for the portion of the asset data **314** into the centralized control database **318**. In one or more embodiments, the metrics engine component **306** determines the one or more metrics for the asset hierarchy based on a model related to a time series mapping of attributes for the asset data **314**. For example, in one or more embodiments, the metrics engine component **306** determines the one or more metrics for the asset hierarchy based on time series mapping of attributes for the asset data **314** with respect to the centralized control database **318**.

(80) In one or more embodiments, in response to the request **320**, the prioritized actions component **326** determines prioritized actions for the portfolio of assets based on attributes for the aggregated data stored in the centralized control database **318**. In an embodiment, the prioritized actions indicate which assets from the portfolio of assets should be serviced first. For example, in an embodiment, the prioritized actions indicate a first asset from the portfolio of assets that should be serviced first, a second asset from the portfolio of assets that should be serviced second, a third asset from the portfolio of assets that should be serviced third, etc. In one or more embodiments, the prioritized actions is a list of prioritized actions for the portfolio of assets based on impact to the portfolio. For instance, in one or more embodiments, the prioritized actions component **326** ranks, based on impact of respective prioritized actions with respect to the portfolio of assets, the prioritized actions to generate the list of the prioritized actions. In one or more embodiments, the prioritized actions component **326** groups the prioritized actions for the portfolio of assets based on relationships, features, and/or attributes between the aggregated data. In one or more embodiments, the prioritized actions component **326** determines the prioritized actions for the portfolio of assets based on a digital twin model associated with one or more assets from the portfolio of assets. Additionally or alternatively, in one or more embodiments, the prioritized actions component **326** determines the prioritized actions for the portfolio of assets based on a digital twin model associated with an operator identity associated with one or more assets from the portfolio of assets.

(81) In one or more embodiments, the prioritized actions component **326** determines the list of the prioritized actions for the portfolio of assets based on metrics associated with the aggregated data. In certain embodiments, in response to the request **320**, the prioritized actions component **326** determines one or more metrics for an asset hierarchy associated with the portfolio of assets. For instance, in one or more embodiments, the prioritized actions component **326** determines one or more metrics for an asset hierarchy associated with the edge devices **161a-161n** in response to the request **320**. In one or more embodiments, the prioritized actions component **326** converts a portion of the asset data **314** into a metric for the portion of the asset data **314** and stores the metric for the portion of the asset data **314** into the centralized control database **318**. In one or more embodiments, the prioritized actions component **326** determines the one or more metrics for the

asset hierarchy based on a model related to a time series mapping of attributes, features, and/or relationships for the asset data **314**. For example, in one or more embodiments, the prioritized actions component **326** determines the one or more metrics for the asset hierarchy based on time series mapping of attributes, features, and/or relationships for the asset data **314** with respect to the centralized control database **318**.

(82) In one or more embodiments, in response to the request **320**, the virtual assistant component **336** performs a natural language query with respect to the voice input data to obtain the one or more attributes associated with the one or more asset insight requests. For example, in one or more embodiments, the virtual assistant component **336** performs natural language processing with respect to the voice input data to obtain the one or more attributes associated with the one or more asset insight requests. In one or more embodiments, the virtual assistant component **336** converts the voice input data into a text string such that the text string associated with one or more textual elements. In one or more embodiments, the virtual assistant component **336** employs natural language processing (e.g., one or more natural language processing techniques) to determine textual data associated with the voice input data. In one or more embodiments, the virtual assistant component **336** queries a natural language database based on the voice input to determine the one or more attributes associated with the one or more asset insight requests. In one or more embodiments, the virtual assistant component **336** provides the one or more attributes, one or more tags, one or more labels, one or more classifications, and/or one or more other inferences with respect to the voice input data. For example, in one or more embodiments, the virtual assistant component **336** performs part-of-speech tagging with respect to the voice input data to obtain the one or more attributes, one or more tags, one or more labels, one or more classifications, and/or one or more other inferences with respect to the voice input data. In one or more embodiments, the virtual assistant component **336** performs one or more natural language processing queries with respect to the centralized control database **318** based on the one or more tags, the one or more labels, the one or more classifications, the one or more attributes, and/or the one or more other inferences with respect to the voice input data.

(83) In one or more embodiments, the virtual assistant component **336** employs one or more machine learning techniques to facilitate determination of the one or more attributes, the one or more tags, the one or more labels, the one or more classifications, and/or the one or more other inferences with respect to the voice input data. For instance, in one or more embodiments, the virtual assistant component **336** performs a fuzzy matching technique with respect to the voice input data to determine the one or more attributes associated with the one or more asset insight requests. Additionally or alternatively, in one or more embodiments, the virtual assistant component **336** provides the voice input data to a neural network model configured for determining the one or more attributes associated with the one or more asset insight requests.

(84) In one or more embodiments, the virtual assistant component **336** obtains aggregated data associated with the portfolio of assets based on the one or more attributes, the one or more labels, the one or more tags, the one or more classifications, /or the one or more other inferences with respect to the voice input data. Additionally, in one or more embodiments, the virtual assistant component **336** determines one or more asset insights for the portfolio of assets based on the aggregated data. In one or more embodiments, the virtual assistant component **336** groups, based on the one or more attributes, the aggregated data based on one or more relationships between assets from the portfolio of assets. In one or more embodiments, the virtual assistant component **336** applies the one or more attributes to at least a first model associated with a first type of asset insight and a second model associated with a second type of asset insight. In one or more embodiments, the virtual assistant component **336** aggregates first output data from the first model and second output data from the second model to determine at least a portion of the aggregated data. In one or more embodiments, in response to the voice input, the virtual assistant component **336** determines prioritized actions for the portfolio of assets based on the one or more attributes. In

certain embodiments, in response to the voice input, the virtual assistant component **336** determines one or more metrics for an asset hierarchy associated with the portfolio of assets. For instance, in one or more embodiments, the virtual assistant component **336** determines one or more metrics for an asset hierarchy associated with the edge devices **161a-161n** in response to the voice input.

(85) In one or more embodiments, in response to the request **320**, the dashboard visualization component **308** generates dashboard visualization data **322** associated with the one or more metrics for the asset hierarchy. For instance, in one or more embodiments, the dashboard visualization component **308** provides the dashboard visualization to an electronic interface of a computing device based on the dashboard visualization data **322**. In one or more embodiments, the dashboard visualization data **322** and/or the dashboard visualization associated with the dashboard visualization data **322** includes the metrics for an asset hierarchy associated with the portfolio of assets. In one or more embodiments, in response to the request **320**, the dashboard visualization component **308** associates aspects of the asset data **314** and/or metrics associated with the asset data **314** stored in the centralized control database **318** to provide the one or more metrics. For example, in one or more embodiment, in response to the voice input, the dashboard visualization component **308** associates aspects of the asset data **314** and/or metrics associated with the asset data **314** stored in the centralized control database **318** to provide the one or more metrics. In an aspect, the dashboard visualization component **308** determines the aspects of the asset data **314** and/or metrics associated with the asset data **314** stored in the centralized control database **318** based on the time series structure and/or the hierarchy structure of asset level of the centralized control database **318**.

(86) In one or more embodiments, the dashboard visualization data **322** and/or the dashboard visualization associated with the dashboard visualization data **322** includes the prioritized actions for the portfolio of assets. In one or more embodiments, the dashboard visualization data **322** and/or the dashboard visualization associated with the dashboard visualization data **322** includes the list of the prioritized actions. In one or more embodiments, the dashboard visualization data **322** and/or the dashboard visualization associated with the dashboard visualization data **322** includes the grouping of the prioritized actions for the portfolio of assets. In one or more embodiments, the dashboard visualization data **322** and/or the dashboard visualization associated with the dashboard visualization data **322** includes the metrics for an asset hierarchy associated with the portfolio of assets.

(87) In one or more embodiments, in response to the voice input, the dashboard visualization component **308** generates the dashboard visualization data **322** associated with the one or more metrics for the asset hierarchy. In one or more embodiments, the dashboard visualization data **322** and/or the dashboard visualization associated with the dashboard visualization data **322** is configured based on the one or more attributes associated with the voice input. In one or more embodiments, the dashboard visualization data **322** and/or the dashboard visualization associated with the dashboard visualization data **322** includes a dashboard visualization element configured to present sensor data related to the portfolio of assets, a dashboard visualization element configured to present control data related to the portfolio of assets, a dashboard visualization element configured to present labor management data related to the portfolio of assets, a dashboard visualization element configured to present warehouse execution data related to the portfolio of assets, a dashboard visualization element configured to present inventory data related to the portfolio of assets, a dashboard visualization element configured to present warehouse management data related to the portfolio of assets, a dashboard visualization element configured to present machine control data related to the portfolio of assets, and/or one or more other dashboard visualization elements associated with the one or more asset insights.

(88) Additionally, in one or more embodiments, the dashboard visualization component **308** performs one or more actions based on the metrics. For instance, in one or more embodiments, the dashboard visualization component **308** generates dashboard visualization data **322** associated with

the one or more actions. In an embodiment, an action includes generating a user-interactive electronic interface that renders a visual representation of the one or more metrics. In another embodiment, an action from the one or more actions includes transmitting, to a computing device, one or more notifications associated with the one or more metrics. In another embodiment, an action from the one or more actions includes providing an optimal process condition for an asset associated with the asset data **314**. For example, in another embodiment, an action from the one or more actions includes adjusting a set-point and/or a schedule for an asset associated with the asset data **314**. In another embodiment, an action from the one or more actions includes one or more corrective action to take for an asset associated with the asset data **314**. In another embodiment, an action from the one or more actions includes providing an optimal maintenance option for an asset associated with the asset data **314**. In another embodiment, an action from the one or more actions includes an action associated with the application services layer **225**, the applications layer **230**, and/or the core services layer **235**.

(89) Additionally, in one or more embodiments, the dashboard visualization component **308** performs one or more actions based on the prioritized actions for the portfolio of assets. In an embodiment, an action includes generating a user-interactive electronic interface that renders a visual representation of the prioritized actions for the portfolio of assets and/or the one or more metrics. In another embodiment, an action from the one or more actions includes transmitting, to a computing device, one or more notifications associated with the prioritized actions for the portfolio of assets and/or the one or more metrics. In one or more embodiments, the dashboard visualization data **322** and/or the dashboard visualization associated with the dashboard visualization data **322** configures the dashboard visualization for remote control of one or more assets from the portfolio of assets based on the one or more attributes associated with the voice input. In one or more embodiments, the dashboard visualization data **322** and/or the dashboard visualization associated with the dashboard visualization data **322** configures a three-dimensional (3D) model of an asset from the portfolio of assets for the dashboard visualization based on the one or more attributes associated with the voice input (e.g., the voice input associated with the request **320**). In one or more embodiments, the dashboard visualization data **322** and/or the dashboard visualization associated with the dashboard visualization data **322** filters one or more events associated with the asset related to the 3D model based on the one or more attributes associated with the voice input. In one or more embodiments, the dashboard visualization data **322** and/or the dashboard visualization associated with the dashboard visualization data **322** configures the dashboard visualization for real-time collaboration between two or more computing devices based on the one or more attributes associated with the voice input.

(90) The system **300** may be configured to interact with or include a computing device **360**. For example, the system **300** may communicatively couple with the computing device **360** via the network **110**. The computing device **360** may be a mobile computing device, a smartphone, a tablet computer, a mobile computer, a desktop computer, a laptop computer, a workstation computer, a wearable device, a virtual reality device, an augmented reality device, or another type of computing device located remote from the asset performance management computer system **302**. In one or more embodiments, the computing device **360** may generate the request **320**. For example, in one or more embodiments, the request **320** may be generated via a visual display (e.g., a user interface) of the computing device **360**. In one or more embodiments, the computing device **360** may generate a voice input. For example, in one or more embodiments, the voice input (e.g., the voice input associated with the request **320**) is generated via one or more microphones of the computing device **360** and/or one or more microphones communicatively coupled to the computing device **360**. In one or more embodiments, the computing device **360** may employ mobile computing, augmented reality, cloud-based computing, IoT technology and/or one or more other technologies to provide performance data, video, audio, text, graphs, charts, real-time data, graphical data, one or more communications, one or more messages, one or more notifications, and/or other media data

associated with the one or more metrics. The computing device **360** includes mechanical components, electrical components, hardware components and/or software components to facilitate determining prioritized actions and/or one or more metrics associated with the asset data **314**. (91) Referring to FIG. **4**, the computing device **360** is shown in greater detail. The computing device **360** may be a part of a system **400** that may include a visual display **404**, one or more speakers **406**, one or more cameras **408**, one or more microphones **410**, a global positioning system (GPS) device **412**, a gyroscope **414**, one or more wireless communication devices **416**, and/or a power supply **418**.

(92) In an embodiment, the visual display **404** is a display that facilitates presentation and/or interaction with one or more portions of the dashboard visualization data **322**. In one or more embodiments, the computing device **360** displays an electronic interface (e.g., a graphical user interface) associated with an asset performance management platform. In one or more embodiments, the visual display **404** is a visual display that renders one or more interactive media elements via a set of pixels. The one or more speakers **406** include one or more integrated speakers that project audio. The one or more cameras **408** include one or more cameras that employ autofocus and/or image stabilization for photo capture and/or real-time video. The one or more microphones **410** include one or more digital microphones that employ active noise cancellation to capture audio data. In one or more embodiments, at least a portion of the voice input is generated via the one or more microphones **410**. The GPS device **412** provides a geographic location for the computing device **360**. The gyroscope **414** provides an orientation for the computing device **360**. The one or more wireless communication devices **416** includes one or more hardware components to provide wireless communication via one or more wireless networking technologies and/or one or more short-wavelength wireless technologies. The power supply **418** is, for example, a power supply and/or a rechargeable battery that provides power to the visual display **404**, the one or more speakers **406**, the one or more cameras **408**, the one or more microphones **410**, the GPS device **412**, the gyroscope **414**, and/or the one or more wireless communication devices **416**. In certain embodiments, the dashboard visualization data **322** associated with the one or more metrics, the prioritized actions and/or the one or more asset insights related to the portfolio of assets is presented via the visual display **404** and/or the one or more speakers **406**.

(93) FIG. **5** illustrates a system **500** that includes an exemplary warehouse and distribution center environment **501**. The system **500** may include components that are connected and monitored by a manager **502** using, for example, the computing device **360**. The manager **502** may be, for example, a shift or area supervisor or a site operations manager and based on the role of the manager **502**, the information displayed to the manager **502** using the systems and devices described herein may change. The manager **502** may be identified to the system **500** using, for example, a user identifier that may describe the managers role associated with access to a dashboard visualization. In some embodiments, particular data may be obtained, aggregated, and displayed to the manager **502** based on the user identifier associated with the manager.

(94) The system **500** may include various components including delivery transportation **505** (e.g., supply chain delivery truck) to load into inventory **508**. An operational control tower **512** may monitor and/or otherwise control operations **510** within the environment **501**. Operations **510** can be performed and/or managed by labor **509**. Operations **510** can include assembly/sorting machines **507**. Once assembled, packaged, and otherwise processed for distribution, delivery transportation **514** (e.g., a freight truck) can be loaded by labor **509** and/or sorting machines **507** and depart for its subsequent destination. Each of the warehouses may be configured such that worker performance is optimized by users and/or automated tools selectively scheduling and assigning tasks and worker equipment, as discussed more particularly below. The system **500** may be configured such that the warehouse environment **501** is capable of generating data regarding the optimized performance of tasks and data associated with key performance indicators (KPIs) such that the manager **502** and other users can measure the performance of each of the various warehouses as explained in greater

detail below. The warehouse environment **501** may be communicatively coupled with one or more other warehouses or systems and the regional manager **502** via a network (e.g., a cloud network such as the network **110**).

(95) FIG. **6** shows a site operations manager **610** using a device **614**. Within an enterprise, the site operations manager **610** may be responsible for analytic thinking that may optimize site performance. For example, the site operations manager may be responsible for analyzing input from a warehouse execution system (WES) and a labor management system (LMS) to maximize throughput and/or productivity of an environment such as the environment **501**. The device **614** may be substantively similar to the computing device **360**. On a hierarchy **612**, a site operations manager may report directly to a DC general manager, who may in turn report to a regional manager. The site operations manager may have various tools at his or her disposal (e.g., the device **614**) to monitor, analyze, and execute actions within the warehouse environment **501**.

(96) The LMS may be used to, for example, measure key worker performance metrics which metrics may be used to determine the effectiveness of individual or groups of workers in the warehouse environment. Worker performance metrics may include, but are not limited to, for example, worker productivity, overall performance against a predetermined plan (e.g., time worked versus number of units moved), “on standard”, percent effectiveness, attendance, engagement (e.g., determined objectively from worker computing devices), clicks per hour by the individual worker, time spent with eyes on a display screen, etc. The LMS may measure work effectiveness in real time and may execute actions and/or make recommendations to a user to optimize labor effectiveness in the warehouse environment. For example, the LMS may execute or recommend where best to locate labor, when to move labor, when to reduce or increase the size of labor, and other factors relating to management of labor. The WES may be used to manage and control functions within the warehouse environment. The WES may control automated facilities, labor, and/or equipment to coordinate each aspect through dynamic optimization based on real-time inputs from the various sensors in the warehouse environment.

(97) FIG. **7** shows a shift supervisor **710** (or “area supervisor”) holding a device **714**. Within an enterprise, the shift supervisor **710** may be responsible for tactical as well as analytical thinking. In a hierarchy **712**, the area supervisor may report to the site operations manager **610**. The shift supervisor **710** may have various tools at his or her disposal (e.g., the device **360** [change this in the Fig]) to monitor, analyze, and execute actions within the warehouse environment **501**. The device **714** may be substantively similar to the computing device **360**.

(98) FIG. **8** depicts an example user interface **810** for of an example computing device **822**. As seen, via user interface **810** one or more aspects of warehouse throughput or productivity can be viewed and one or more tasks can be executed and/or assigned to one or more users (e.g., crew member) or equipment. The example user interface **810** may display notifications related to throughput or performance or the one or more tasks, such as, for example, a newly assigned task or feedback related to an already-assigned task, which may include information controls for users to accept, snooze, and/or otherwise interact with information displayed (e.g., propose or execute modifications to a task, work plan, and/or the like).

(99) FIG. **9** shows an exemplary dashboard **900** for monitoring, analyzing, and executing one or more warehouse actions. The dashboard **900** may be displayed for instance, on the display of the example computing device **822** or the computing device **360**. The dashboard **900** may include information such as analytics related to various aspects of the warehouse environment **901** and may make one or more recommendations based on data generated by the one or more sensors in the warehouse environment **901**. The dashboard can display, for example, information in a machine related to the machines **910**, the labor **915**, and the inventory **912** within the warehouse environment **501**. The dashboard **900** may display, for example, one or more operational descriptors of the various aspects of the warehouse. For example, the dashboard may display asset health, performance, quality, a functional status (e.g., running, not running, degraded __%, etc.), a

capacity, a maintenance status (e.g., maintenance requested, last maintenance completion, lists of maintenance, etc.), a list of users and authorized users, and other information related to the particular aspect chosen to be displayed (e.g., the machines **910**).

(100) In some embodiments, the dashboard **900** may be configured to provide access to the WES. For example, a user may be able to see order backlog at different functional areas (e.g., order picking, shipping, routing, etc.) The dashboard **900** may provide real-time access to order receipt, processing, and throughput to properly track order waves.

(101) In some embodiments, the dashboard **900** may be configured to display information generated in the LMS such that the user of the dashboard is integrated into the LMS through the dashboard **900**. The LMS may display information such as, for example, a current allocation of labor throughout the warehouse, or the particular skills and qualifications of individual warehouse associates. The dashboard **900** may be configured to display recommendations regarding where the best location is to place workforce in order to maximize operations. The system may develop and track benchmarks (e.g., maximum work productivity) in order that worker productivity on any given day can be measured against maximum work productivity. In some embodiments, the dashboard **900** may be configured such that recommendations regarding capacity smoothing are provided to a user. The capacity smoothing recommendations may be based on, for example, a comparison work capacity on any particular day at a given work location versus the maximum capacity for work at that location.

(102) In some embodiments, the dashboard **900** may display one or more insights or recommendations with respect to the integration of worker tasks, which may be integrated using, for example, one or more audio devices **914** (which may be configured to play and sense audible data (e.g., using speakers and/or a microphone)) and one or more computing devices **916** (which may include a display and an input/output device (e.g., a touchscreen)).

(103) In some embodiments, the visualization may be configured based on a user identifier associated with a user of the computing device used to display the dashboard **900** (e.g., the manager **502**). For example, the manager **502** may receive different information based on his or her user profile as compared with a regional manager or other level of user of the system (as shown in the hierarchies herein). The user profile may be determined based on one or more hierarchical structures such as the hierarchies shown in FIGS. **6** and **7**. The user identifier may be used to configure a display (e.g., present different information) to a user based on the identity of the user.

(104) FIG. **10** illustrates an exemplary electronic interface **1000** according to one or more embodiments of the disclosure. In an embodiment, the electronic interface **1000** is an electronic interface of the computing device **360** that is presented via the visual display **404**. In one or more embodiments, a dashboard visualization is presented via the electronic interface **1000**. In certain embodiments, the data visualization presented via the electronic interface **1000** presents one or more asset insights **1002** and/or one or more notifications **1004** via the dashboard visualization associated with the electronic interface **1000**. The asset insights **1002** and/or notifications **1004** may be related to one or more features of the warehouse system **500** of FIG. **5** (e.g., the machines **522**, the labor **518**, etc.) In one or more embodiments, the data visualization presented via the electronic interface **1000** includes a 3D model **1006** associated with an asset from a portfolio of assets. In one or more embodiments, one or more events associated with the 3D model **1006** can be filtered and/or information associated with the one or more events can be displayed in response to selection of one or more interactive buttons associated with the 3D model **1006**.

(105) FIG. **11** illustrates a schematic view of a material handling system **1100**, in accordance with one or more embodiments described herein. In one or more embodiments, the material handling system **1100** corresponds to an enterprise from the enterprises **160a-n**. In one or more embodiments, the material handling system **1100** includes one or more assets from a portfolio of assets. The material handling system **1100** includes at least one vision system **1102** with one or more LiDAR based sensors **1104**, according to an example embodiment. The material handling

system **1100** may correspond to a material handling environment for example, but not limited to, a distribution center, a shipping station, a warehouse, an inventory, etc. According to some example embodiments, the material handling system **1100** includes one or more conveyors for handling various items such as, cartons, totes, shipping packages, boxes etc. As illustrated, the material handling system **1100** includes a sorter portion **1106** for selectively identifying, sorting and/or diverting one or more articles **1108** to one of the destinations **1110**, such as, but not limited to, takeaway conveyors, chutes, and the like. In some examples, the diverted articles may be sent to shipping **1112** for shipping to a destination, for example, a store. While the example as shown in FIG. **11** may illustrate a paddle sorter, it is noted that the scope of the present disclosure is not limited to a paddle sorter. In some examples, the material handling system **1100** may include other types of sorter(s) may be implemented, including, but not limited to, pusher/puller sorters, pop-up transfer sorters, and/or cross-belt sorters.

(106) Although the LiDAR sensors **1104** are illustrated to be located within the vision system **1102**, however, according to various example embodiments described herein, multiple LiDAR based sensors are installed at various sections of the material handling system **1100**. In other words, the LiDAR sensors **1104** may be positioned at various different sections (e.g. workstations) within the material handling system **1100**. Further, in one or more embodiments, these LiDAR based sensors are communicatively coupled (e.g. remotely connected) to the vision system **1102**, via a communication network (e.g. wireless or wired network).

(107) Referring to FIG. **11**, illustratively, a first LiDAR sensor unit **1104-1** is installed near an area corresponding to an automated storage and retrieval system (ASRS) **1122**. Similarly, a second LiDAR sensor unit **1104-2** may be installed near another area corresponding to a singulation system along the sorter **1106**. In another example, similar LiDAR based sensor units may be located at the shipping station **1112** or at various other positions (not shown) along the sorter **1106**. Accordingly, the material handling system **1100** may include many more such LiDAR sensor units that are installed or mounted at various sections (e.g. dedicated zones) of a material handling environment. As stated before, in one or more embodiments, these sensor units are communicatively coupled to the vision system **1102**, via the communication network. These LiDAR based sensor units may be capable of capturing a data stream (e.g. 3D data stream) representative of a 3D scan of that area where the respective LiDAR sensor unit is located. In one or more embodiments, the data stream is used by the vision system **1102** to monitor, one or more articles **1114**, machines, and/or workers present in various sections of the material handling system **1100**.

(108) As illustrated, in one or more embodiments, the material handling system **1100** includes a sorter portion (e.g. the sorter **1106**) that receives the one or more articles **1114** from an induction portion **1116**. In some examples, the induction portion **1116** is associated with a singulation system **1118** that is configured to generate spacing between the one or more articles **1114**. For example, the induction portion **1116** may comprise various mechanical components e.g. configurations of belt units and/or mechanical actuators with end effectors, which may create the required spacing between the one or more articles **1114**. In accordance with some example embodiments, LiDAR based sensors of the LiDAR sensor unit **1104-2** may capture a 3D scan of various operations and/or activities that may be performed on the singulation system **1118**.

(109) In some examples, the induction portion **1116** receives articles **1114** from a merge portion **1120**, as shown in FIG. **11**. The merge portion **1120** may have multiple accumulation lanes and/or conveyors for releasing articles in a slug and/or zipper fashion onto the induction portion **1116**. In some examples, the merge portion **1120** may receive the one or more articles **1114** from a receiving system and/or an automated storage and retrieval system (ASRS) **1122**. Additionally, or alternatively, the merge portion **1120** may receive the one or more articles from other sources. In some example embodiments, the ASRS **1122** may also include a separate vision system (VS1) **1124** with one or more LiDAR based sensor units (similar to **1104-1**, **1104-2**) that may be installed at

various locations within the ASRS **1122**.

(110) According to some example embodiments, the LiDAR sensors **1104** of the vision system **1102** are configured for scanning a target area of the material handling environment and generate one or more data streams. In some example embodiments, a processor of the vision system **1102** may utilize a data stream to construct 3D point cloud that may represent a 3D-scan of the target area. As an example, a data stream recorded by these LiDAR sensors may capture various operations of a material handling site e.g. movement of the one or more articles **1114**, e.g. from the induction portion **1116** towards the sorter portion **1106** or from the ASRS **1122** to the merge portion **1120**, and so on. Further, data streams from various LiDAR sensors **1104** may also capture operations and/or actions performed by various machines of the material handling site. For instance, in an example, the data stream may capture movement of various mechanical components e.g. conveyor belts etc. of the singulation system. Furthermore, the data streams may also capture operations performed by one or more workers in that target area.

(111) According to some example embodiments, one or more components of the example material handling system **1100**, such as, but not limited to, the sorter portion **1106**, the induction portion **1116**, the merge portion **1120**, the vision system **1102**, and/or the like, may be communicably coupled to at least one of a central system e.g., a distribution center (DC) execution system **1126** (or a warehouse execution system, a warehouse management system, a labor management system, a machine control system, and/or another system) and/or a controller **1128**. In one or more embodiments, the controller **1128** is configured for machine control. The term “communicably coupled” refers to two or more components (for example, but not limited to, the sorter portion **1106**, the induction portion **1116**, the merge portion **1120**, the vision system **1102**, the DC execution system **1126** and the controller **1128** as shown in FIG. **11**) being connected through wired means (for example but not limited to, wired Ethernet) and/or wireless means (for example but not limited to, Wi-Fi, Bluetooth, ZigBee), such that data and/or information may be transmitted to and/or received from these components.

(112) FIG. **12** illustrates a schematic view **1200** of a target area of the material handling system **1100** including the LiDAR based vision system, according to an example embodiment. The target area may correspond to an area of a distribution center (DC). In one or more embodiments, the DC may receive goods in bulk from various manufacturers, suppliers, stores, shipping stations, and the like, and may store and/or handle received goods until the goods are further picked and/or shipped. Further, the received goods may be transferred into totes and/or containers of appropriate size, shape, material, etc. for storage and/or further processing. In accordance with some example embodiments described herein, the DC may include a vision system **1201** that may be communicatively coupled, via a network **1203**, to multiple LiDAR based sensor units VS1, VS2, VS3, VS4, etc., as illustrated in FIG. **12**. Similar to as described earlier in reference to FIG. **11**, these LiDAR based sensor units (VS1-VSn) may be capable of recording data streams including 3D scan of a target area. The network **1203** may correspond to a wired or wireless communication network. In one or more embodiments, the vision system **101** corresponds to an asset from a portfolio of assets.

(113) Illustratively, in some example embodiments, the DC may have a replenishment area **1202** for replenishing one or more containers **1204** with goods arriving at the replenishment area **1202** in multiple stock keeping units (SKUs) **1206**. The term ‘replenishment area’ as used herein may refer to an area, system, workstation, and the like in the DC for transferring goods from the multiple SKUs **1206** into one or more containers **1204**. The replenishment area **1202** may have a collaborative system of multiple material handling devices and systems, such as, but not limited to, infeed conveyors, outfeed conveyors, goods to operator workstations, devices, staging units, and the like. Transferring goods from an SKU into the containers **1204** may be automated, for example, may be done by a robotic tool, and/or may be a manual process carried out by an operator, such as operators **1208** and **1210**, as shown in FIG. **12**. In accordance with some example embodiments

described herein, one or more LiDAR based sensors are associated with the replenishment area **1202** to perform a 3D scan that captures activities, operations, devices, and/or workers in the replenishment area **1202**. Accordingly, in one or more embodiments, there are multiple vision systems that may be associated with different sections of the DC. In one or more embodiments, these vision systems employ LiDAR based sensors to record the activities related to operators, items, and/or machines within the respective section. As an example, as illustrated in FIG. **12**, a vision system unit VS2 with one or more LiDAR sensors **204** is associated with the replenishment area **1202**.

(114) According to said example embodiments, an SKU **1206** may include goods of a similar type, size, shape, and/or any other common characteristic. In an embodiment, one or more SKUs **1206** may be grouped together and stacked on a pallet **1212**, as shown in FIG. **12**. The SKUs **1206** may be grouped based on a common characteristic, such as type of goods. Additionally, or alternatively, mixed SKUs **1206** may be grouped randomly and placed on the pallet **1212**. The SKUs **1206** may be grouped and stacked on the pallet **1212** at the DC for ease of handling. In some embodiments, each SKU **1206** and each pallet **1212** may include a respective identifier (e.g. a barcode label, RFID tag) that is scanned at the replenishment area **1202**. The scanned information indicates, in one or more embodiments, a location of the pallet **1212** at the replenishment area **1202**. In some example embodiments, one or more LiDAR based sensor units may also be located in the DC to perform 3D scan of an area including the SKUs **1206** and/or pallets **1212**. Illustratively, in an example, two vision system units VS1 and VS4 with LiDAR sensors may be located to track activities, operations, and/or characteristics associated with the SKUs **1206** and/or the pallets **1212**.

(115) In accordance with one or more embodiments, the replenishment area **1202** includes a gravity flow rack **1214** for staging and/or conveying one or more containers **1204**. Further, the replenishment area **1202** may include multiple replenishment zones. The gravity flow rack **1214** may be placed between different replenishment zones, such that the gravity flow rack **1214** may convey replenished containers from a first replenishment zone **1216** to a second replenishment zone **1218** and convey empty containers back from the second replenishment zone **1218** to the first replenishment zone **1216**. The gravity flow rack **1214** may also function as a staging area for the empty and/or filled containers **1204** until the containers **1204** are handled by the operator **1208** and/or a robotic tool. In accordance with some example embodiments, the vision system unit VS2 may scan the area including the gravity flow rack **1214**.

(116) The replenishment area **1202** may further include one or more devices **1220**. The devices **1220** may refer to any portable and/or fixed device (e.g. a human machine interface HMI) that may be communicably coupled to a central controller (e.g. the controller **1128**). In some examples, the devices **1220** may include an input/output interface which may be used for assisting the operator **1208** in the replenishment process. According one or more embodiments, the devices **1220** correspond to or include for example, but not limited to, scanners, imagers, displays, computers, communication devices, headsets, and the like. According to some example embodiments, the devices **1220** may further receive data, commands, workflows, etc. from the central controller and/or any other device that may be communicably coupled to the devices **1220**. According to some example embodiments, the vision system units VS1 and VS5 using the LiDAR based sensors may perform a 3D scan of area including the one or more devices **1220**.

(117) According to some example embodiments, the data stream captured by the vision system **1201** may monitor various activities, operations, individuals, and/or equipment in the DC. For instance, the data stream may be used to monitor arrival of the pallets **1212** having one or more SKUs **1206** at the replenishment area **1202** of the DC. Further, the data stream may monitor scanning of a pallet identifier and/or an SKU identifier using the devices **1220** by any of the operators **1208** and/or **1210**. In some example embodiments, the data stream captured by the LiDAR sensors **204** of the vision system **1201** may also include an operation by a robotic tool (not shown) and/or the operators (**1208**, **1210**) to pick one or more of the containers **1204** on the gravity

flow rack **1214** for replenishing the one or more containers **1204** with the goods that may be in the SKU **1206** and/or the pallet **1212**. Further, in some example embodiments, the data stream captured by the LiDAR sensors **204** of the vision system units VS2, VS3, and/or VS4 may include conveyance or movement of the one or more containers **1204** that may be on the gravity flow rack **1214**. In this aspect, the containers **1204** may be conveyed from the first replenishment zone **1216** to the second replenishment zone **1218** through the gravity flow rack **1214**. In some example embodiments, the data stream may also include monitoring of empty container(s) that may be placed on the gravity flow rack **1214** for transferring back to the first replenishment zone **1216** for receiving goods from a next SKU and/or pallet. In an example embodiment, the data stream also includes movement of some containers to one or more shuttle totes that can be moved for storing goods in an Automated Storage and Retrieval System (ASRS) in the DC.

(118) FIG. **13** illustrates an example scenario **1300** depicting monitoring of an operation performed by a worker in a material handling environment by using LiDAR based vision system (e.g. the vision system **1102**), according to an example embodiment. In some example embodiments, the operation may be performed in a replenishment zone of a distribution center. FIG. **13** illustrates an example of a replenishment zone **1302** of a distribution center. As described earlier, in one or more embodiments, a material handling environment includes a plurality of vision systems. Illustratively, in some example embodiments, a distribution center DC includes a plurality of vision systems (**1301**, **1303**, **1307** etc.). Each of these vision systems (**1301-1307**) include one or more LiDAR based sensors that may be installed and/or mounted at various sections of the material handling environment. In this aspect, each of these vision systems **1301-1307** are capable of capturing a data stream (i.e. a 3D scan) of a target area. In one or more embodiments, the vision systems **1301-1307** correspond to respective assets from a portfolio of assets.

(119) According to some example embodiments, the operation monitored by using LiDAR based vision systems corresponds to replenishing of one or more containers. The containers may be placed on a gravity flow rack **1308** and, in one or more embodiments, is replenished with goods from the one or more SKUs **1310** that may be arriving at a replenishment area of the replenishment zone **1302**. According to some example embodiments, there may be different sizes of containers for replenishment in the DC. For instance, a first set of containers **1304** may be of moderate size, whereas a second set of containers **1306** may be smaller than the first set of containers **1304**, and a third set of containers **1305** may be larger than containers of the first set of containers **1304**. In one or more embodiments, the replenishment of containers is based on a size of the containers. According to one or more embodiments, each of the containers **1304**, **1306**, **1305** have an associated container identifier (not shown). The container identifier may refer to a unique identifier that may be used to identify a particular container, such as, but not limited to, a serial number, a barcode label, RFID tag, etc. The container identifier may include information regarding the container, such as, but not limited to, type, size, capacity, weight, shape, and the like.

(120) In accordance with said example embodiments, a container identifier for a container may be scanned before performing each replenishment operation for that container. By scanning the container identifier, a central controller (e.g. the controller **1128**) and/or any other computing device in the DC, may track an occupied volume of the container. Further, based on this information, the central controller may calculate a current capacity i.e. based on a maximum capacity of the container and the occupied volume. Said that, in accordance with said example embodiments, to maximize storage capacity and overall efficiency, it may be desired to pick appropriately sized container(s) from various sized containers for storing goods from the SKUs **1310**.

(121) FIG. **14** illustrates another example scenario **1400** depicting another operation performed in a material handling environment that is monitored by using LiDAR based vision system (e.g. the vision system **1102**), according to an example embodiment. FIG. **14** illustrates a perspective view of a second replenishment zone **1402** of the distribution center (DC), in accordance with one or

more embodiments of the present disclosure. Illustratively, in some example embodiments, a distribution center DC includes a plurality of vision systems (**1401**, **1403**, **1405** etc.). Each of these vision systems (**1401-1405**) includes one or more LiDAR based sensors that may be installed and/or mounted at various sections of the material handling environment. In this aspect, each of these vision systems **1401-1405** is configured to capture a data stream (i.e. a 3D scan) of a target area. In one or more embodiments, the plurality of vision systems (**1401-1405**) correspond to respective assets from a portfolio of assets. In accordance with some example embodiments, the data stream from the LiDAR sensor-based vision system captures an operation related to a replenishment process in the second replenishment zone **1402**.

(122) According to some example embodiments, a replenishment process illustrated in FIG. **14** includes replenishing of one or more containers from a second set of containers **1404** with goods from the replenished first set of containers **1406** that may be arriving at the second replenishment zone **1402** (e.g. through the gravity flow rack **1408**). In some example embodiments, the second set of containers **1404** may correspond to shuttle totes used in an ASRS (e.g., the ASRS **1122**) that may be having multiple compartments of different size. The shuttle totes may be partially filled or empty and may be used to store goods in a storage facility, such as the ASRS **1122** as illustrated in FIG. **11**.

(123) FIG. **15** depicts an example system **1500** that may execute techniques presented herein. FIG. **15** is a simplified functional block diagram of a computer that may be configured to execute techniques described herein, according to exemplary embodiments of the present disclosure. Specifically, the computer (or “platform” as it may not be a single physical computer infrastructure) may include a data communication interface **1560** for packet data communication. The platform also may include a central processing unit (“CPU”) **1520**, in the form of one or more processors, for executing program instructions. The platform may include an internal communication bus **1510**, and the platform also may include a program storage and/or a data storage for various data files to be processed and/or communicated by the platform such as ROM **1530** and RAM **1540**, although the system **1500** may receive programming and data via network communications. The system **1500** also may include input and output ports **1550** to connect with input and output devices such as keyboards, mice, touchscreens, monitors, displays, etc. Of course, the various system functions may be implemented in a distributed fashion on a number of similar platforms, to distribute the processing load. Alternatively, the systems may be implemented by appropriate programming of one computer hardware platform.

(124) The systems and methods of this disclosure can be cloud-based, multi-tenant solutions configured to deliver optimized work instructions tailored for specific vertical workflows utilizing an easy to deploy, scalable, and configurable data model and software suite to deliver performance insights and improve worker productivity.

(125) It should now be understood that current tools for monitoring warehouse operations may require users to extract data and generate insights from separate systems with disconnected interfaces and non-communicating components but that disaggregated, stovepiped systems may be overcome using systems and methods of receiving one or more edge devices to generate a dashboard visualization including one or more actions or recommendations, which dashboard visualization can simultaneously be used to execute one or more actions or follow the generated recommendation, thus overcoming the disaggregated and stovepiped nature of the various systems generating and sending data.

(126) Other embodiments of the disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. It is intended that the specification and examples be considered as exemplary only, with a true scope and spirit of the invention being indicated by the following claims.

Claims

1. A system for monitoring performance of a warehouse, comprising: one or more processors; a memory, comprising one or more programs stored in the memory, the one or more programs comprising instructions configured to: receive a voice input, wherein the voice input comprises a request to generate a dashboard visualization associated with one or more edge devices, the request comprising: a categorization of one or more of the one or more edge devices; and at least one operational descriptor; and in response to the request: process the voice input, wherein processing the voice input comprises: converting the categorization of the voice input into a first text string and the at least one operational descriptor of the voice input into a second text string; determining one or more attribute tags associated with the voice input by querying a natural language database using at least one of the first text string or the second text string; and performing, using at least one of the one or more attribute tags, a natural language query to obtain aggregated data from a centralized control database; determine a status associated with the one or more edge devices for the aggregated data; configure the dashboard visualization to a remote-control format such that the dashboard visualization is configured to remotely control a first edge device of the one or more edge devices and enable real-time collaboration with a remote computing device; provide the dashboard visualization to an electronic interface of a computing device, the dashboard visualization comprising the status associated with the one or more edge devices; determine a list of prioritized actions for the one or more edge devices based on the status; and optimize a status of the one or more edge devices based on the determined list of prioritized actions.
2. The system of claim 1, the request further comprising a user identifier, the user identifier describing a user role for a user associated with access of the dashboard visualization via the electronic interface, and, in response to the request, the aggregated data is obtained based on the user identifier.
3. The system of claim 2, the one or more programs further comprising instructions configured to: configure the dashboard visualization based on the user identifier.
4. The system of claim 1, the one or more programs further comprising instructions configured to: provide the list of prioritized actions to a user via the dashboard visualization.
5. The system of claim 4, the one or more programs further comprising instructions configured to: group the prioritized actions for the one or more edge devices based on relationships between the aggregated data; and configure the dashboard visualization based on the grouping of the prioritized actions for the one or more edge devices.
6. The system of claim 4, the one or more programs further comprising instructions configured to: rank, based on impact of respective prioritized actions with respect to the one or more edge devices, the prioritized actions to generate the list of the prioritized actions; and provide the ranked list of the prioritized actions to the user via the dashboard visualization.
7. The system of claim 1, the one or more programs further comprising instructions configured to: determine one or more recommendations for the one or more edge devices based on the status; and provide the one or more recommendations to a user via the dashboard visualization.
8. The system of claim 1, the one or more programs further comprising instructions configured to: configure the dashboard visualization to provide a visualization of operational capacity of the one or more edge devices.
9. The system of claim 1, the one or more programs further comprising instructions configured to: receive an input from a labor management system; receive an input from a warehouse execution system, wherein the dashboard visualization is based on a status associated with one or more of the labor management system and the warehouse execution system.
10. The system of claim 9, the one or more programs further comprising instructions configured to: receive, from the labor management system, a plurality of worker performance metrics based on

data from the labor management system, wherein the dashboard visualization is based on the plurality of worker performance metrics.

11. The system of claim 1, wherein to optimize the status of the one or more edge devices based on the determined list of prioritized actions the one or more programs further comprise instructions configured to perform a first action, wherein the first action comprises adjusting a set-point associated with at least one edge device of the one or more edge devices.

12. The system of claim 1, wherein processing the voice input is performed using a virtual assistant and providing the dashboard visualization is performed using a dashboard visualization component.

13. A method of monitoring performance of one or more a warehouses comprising: receiving a voice input, wherein the voice input comprises a request to generate a dashboard visualization associated with one or more edge devices, the request comprising: a categorization of one or more of the one or more edge devices; and at least one operational descriptor; and in response to the request: processing the voice input, wherein processing the voice input comprises: converting the categorization of the voice input into a first text string and the at least one operational descriptor of the voice input into a second text string; determining one or more attribute tags associated with the voice input by querying a natural language database using at least one of the first text string or the second text string; and performing, using at least one of the one or more attribute tags, a natural language query to obtain aggregated data from a centralized control database; determining a status associated with the one or more edge devices for the aggregated data; configuring the dashboard visualization to a remote-control format such that the dashboard visualization is configured to remotely control a first edge device of the one or more edge devices and enable real-time collaboration with a remote computing device; and providing the dashboard visualization to an electronic interface of a computing device, the dashboard visualization comprising the status associated with the one or more edge devices.

14. The method of claim 13, wherein one or more of the one or more edge devices that generates the obtained aggregated data is capable of encrypting the obtained aggregated data.

15. The method of claim 13, further comprising: determine a list of prioritized actions for the one or more edge devices based on the status; and provide the list of prioritized actions to a user via the dashboard visualization.

16. The method of claim 15, further comprising: grouping prioritized actions for the one or more edge devices based on relationships between the aggregated data; and configuring the dashboard visualization based on the grouping of the prioritized actions for the one or more edge devices.

17. A system comprising: at least one warehouse, each at least one warehouse including a one or more edge devices configured to access a network; and a computing device comprising: a display; one or more processors; a memory including one or more programs stored therein, which, when executed by the one or more processors, cause the computing device to: receive a voice input, wherein the voice input comprises a request to generate a dashboard visualization associated with the one or more edge devices, the request comprising a categorization of the one or more edge devices and at least one operational descriptor; process the voice input wherein processing the voice input comprises: converting the categorization of the voice input into a first text string and the at least one operational descriptor of the voice input into a second text string; determining one or more attribute tags associated with the voice input by querying a natural language database using at least one of the first text string or the second text string; and performing, using at least one of the one or more attribute tags, a natural language query to obtain aggregated data from a centralized control database; determine prioritized actions associated with the one or more edge devices for the aggregated data; configure the dashboard visualization to a remote-control format such that the dashboard visualization is configured to remotely control a first edge device of the one or more edge devices and enable real-time collaboration with a remote computing device; provide the dashboard visualization to the display, the dashboard visualization comprising the prioritized actions associated with a portfolio of assets, determine a list of prioritized actions for the one or

more edge devices based on a status; and optimize the status of the one or more edge devices based on the determined list of prioritized actions, wherein the visualization is based on one or more of: the categorization of one or more of the edge devices; and the at least one operational descriptor.
